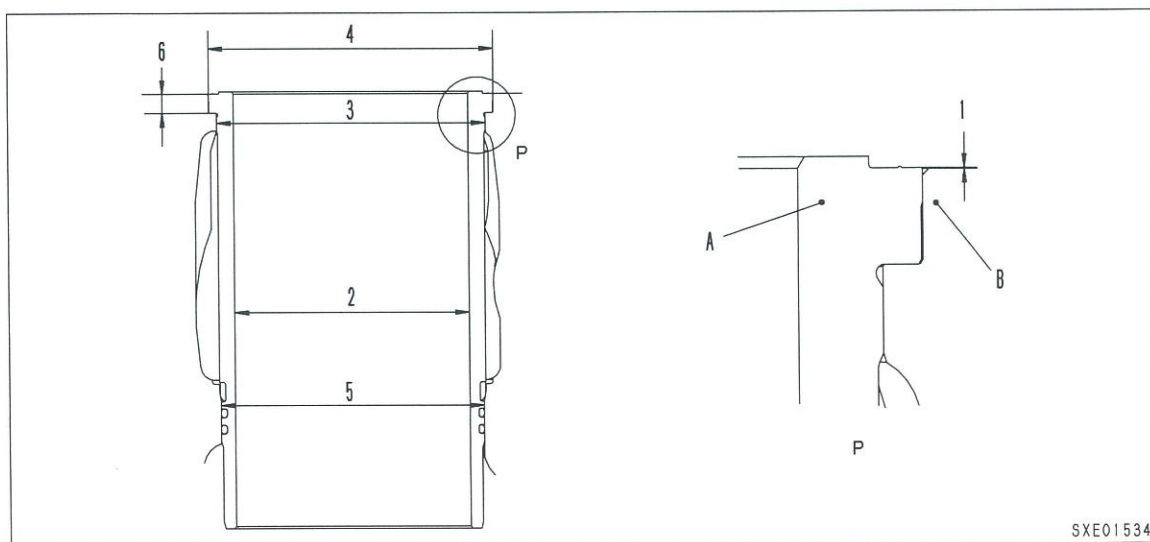


Cylinder liner



- A. Cylinder liner section
B. Cylinder block section

Unit: mm

No.	Check item		Criteria		Remedy
1	Projection of cylinder liner		Permissible range: 0.07 – 0.15 Difference between cylinders: Max. 0.05		Replace cylinder liner or cylinder block
2	Inside diameter of cylinder liner		Standard size	Tolerance	Repair limit
			170	+0.040 0	170.24
		Roundness of inside	0.020		0.08
		Cylindricality of inside	0.020		0.08
3	Outside diameter of cylinder liner (at bottom of counter-bore portion)	Standard size		Tolerance	
		194.5		194.565 – 194.615	
	Interference of cylinder liner and block (at bottom of counter-bore portion)	Standard interference: 0.025 – 0.135			
4	Outside diameter of cylinder liner (at counterbore portion)	Standard size		Tolerance	
		206		205.965 – 206.015	
	Interference of cylinder liner and block (at counterbore portion)	Standard clearance		Clearance limit	
		Interference: 0.85 to Clearance: 0.025		—	
5	Outside diameter of cylinder liner (at O-ring)	Standard size		Tolerance	
		190.40		190.24 – 190.29	
	Clearance between cylinder liner and block (at O-ring)	Standard clearance		Clearance limit	
		0.05 – 0.16		—	
6	Dispersion of depth of counter-bore	Standard		Repair limit	
		—		0.05	
					Correct by grinding



ENGINE RECEIVING INSPECTION CHECK SHEET

KOMATSU ENGINE 6D170E-5

MACHINE MODEL

D 375-6

WONO:

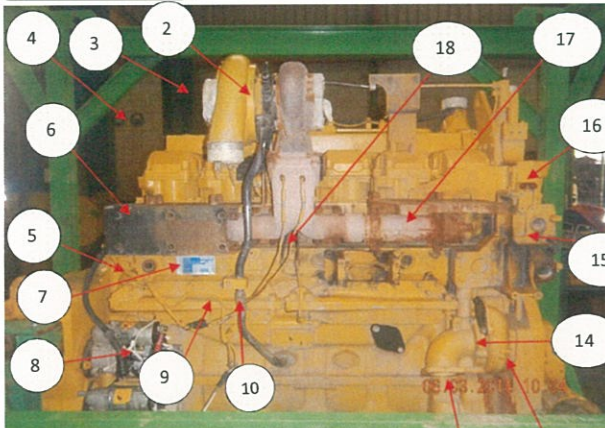
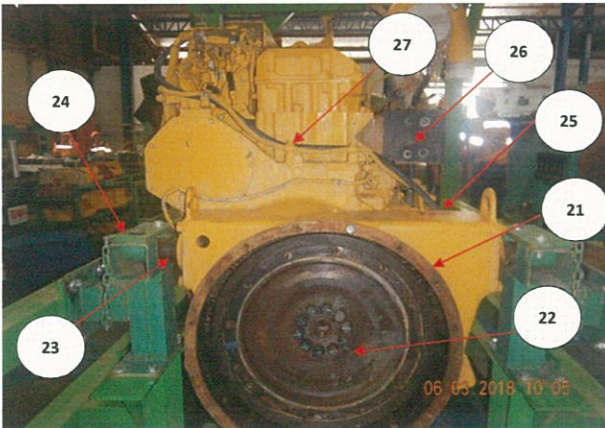
DISASS'Y SECTION

**RECEIVING
INSPECTION SHEET
ENGINE**

MACHINE MODEL	D375-6R	ENGINE PART NO.	6240-B0-0040	HOUR METER	
ENGINE MODEL	SAA6D170E-5	WO NO.		LIFETIME	
ENGINE SERIAL NO		RECEIVING DATE		REVISION NO.	01
HISTORICAL		SITE		PAGE	1 OF 2

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

SKETCH DRAWING OR PHOTOS	NO.	ITEM	INSPECTION CONDITION
Right Side View    Rear Side View	1	Painting (Y / B)	
	2	Turbocharger (P/N:6502-51-5020) S/N: 01184	
	3	Tube support	
	4	Hose Air cleaner	
	5	Water temperature Sensor Assy	
	6	Cover safety Exhaust Manifold (heat Shiled Assy)	
	7	Name plate engine	
	8	Starting motor (2pcs) Upper S/N : Lower S/N :	
	9	Oil cooler (2pcs)	
	10	Tube drain Turbocharger	
	11	Oil pan	
	12	Valve water Drain Cock	
	13	Water inlet cover plate	
	14	Water pump assy	
	15	Tube Exhaust Gas EGR Re-Circulation	
	16	Thermostat Housing	
	17	Exhaust manifold	
	18	Exhaust temperature sensor (2pcs)	
	19	Air cleaner	
	20	Bracket Air Cleaner	
	21	Flywheel Housing	
	22	Flywheel	
	23	Rear Bracket	
	24	Engine stand	
	25	Sensor Revolution	
	26	Heat Tab position	
	27	Main harness	

Notes :

RECEIVING INSPECTION SHEET ENGINE

MACHINE MODEL	D375-6R	ENGINE PART NO.	6240-B0-0040	HOUR METER	
ENGINE MODEL		WO NO.		LIFETIME	
ENGINE SERIAL NO		RECEIVING DATE		REVISION NO.	01
HISTORICAL		SITE		PAGE	1 OF 2

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

SKETCH DRAWING OR PHOTOS	NO.	ITEM	INSPECTION CONDITION
Left Side View Rear Side View	28	Oil pressure sensor Ass'y	
	29	Oil Temperature sensor ass'y	
	30	Oil Filter	
	31	Tube Air	
	32	Air Connector Intake manifold	
	33	Alternator Assy (90) (P/N :600-861-9120)	
		P/N :	
	34	Bracket Belt drive Guard	
	35	AC Compressor Assy	
	36	Air Compressor Assy (P/N : 6210-81-3122)	
		S/N :	
	37	Corrosion resistor	
	38	Blowby Sensor	
	39	Gauge Oil (STD : H+ 0 ~ 10 mm)	
		Act : H +.....10.....mm	
	40	Tube Oil Filter Cap	
	41	Oil Level sensor	
	42	Drain Plug With Valve	
	43	Priming Pump Assy With Cover	
	44	Tube Inlet	
	45	Tube Outlet	
	46	Engine Controller Assy	
		S/N : 37107596	
		ESN : 10000065	
		Assy P/N : 60046395003	
	47	Fuel Pre Filter	
	48	Fuel Supply Pump Assy (P/N : 6245-71-1100)	
		S/N : BLANK	
	49	Feed Pump	
	50	Dust Indicator	
	51	Electrical Wiring Harness	
	52	AIR Bleed Valve	
	53	Fuel Filter Assy	
	54	Front Hanger plate	
	55	Tube Water Outlet	
	56	Cooler mounting	
	57	Tube Water	
	58	Block Water Housing	
	59	Damper Assy	
	60	Front Suport	
	61	Oil Pump Assy	
	62	Alternator Drive pulley	
	63	Tension Pulley Assy	
	64	V-belt Alternator	

Notes :

Inspected by,

Approved by,

REV.01



PT SAPTAINDRA SEJATI

PLANT REBUILD CENTER

ENGINE DISASSEMBLY

CHECK SHEET - 2

KOMATSU ENGINE SAA6D170E-5 SERIES

MACHINE MODEL

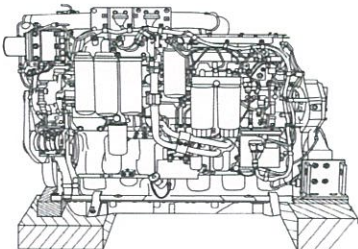
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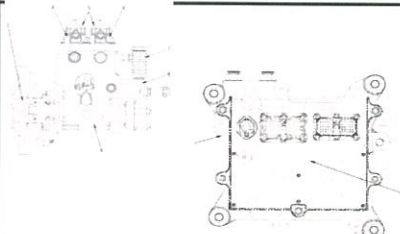
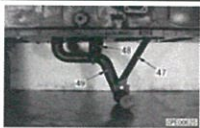
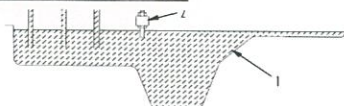
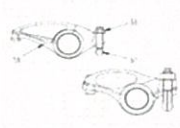
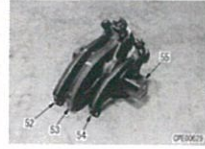
ENGINE SECTION

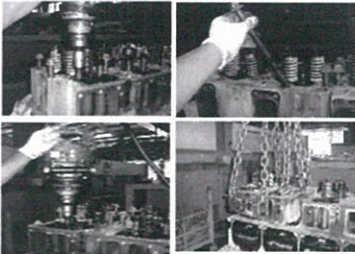
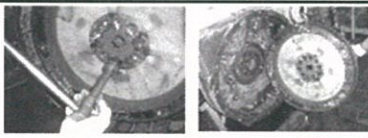
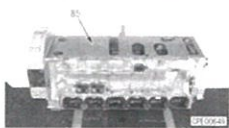
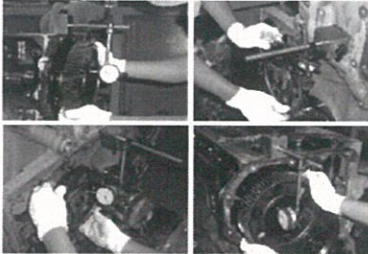
KOMATSU ENGINE 6D170E-5

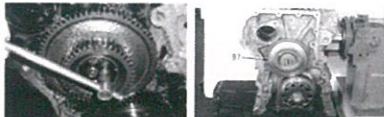
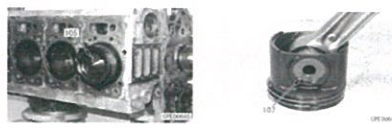
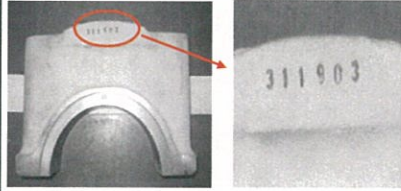
JOB NUMBER		DOCUMENT NO.	RC/QR02/EG/01/03
ENGINE MODEL	SAA6D170E-5	PUBL. DATE	09/08/2010
ENGINE SER. NO		REVISION NO.	01
MACHINE MODEL		PAGE	

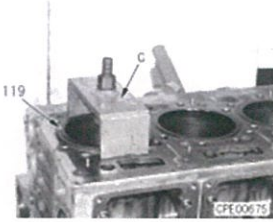
PROCESS	START		DISASSEMBLY BY	
	FINISH			

NO	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REUSE	SALVAGE	REPLACE	SKETCH DRAWING OR PHOTOES	REMARKS									
1	Engine ass'y	- Setting engine to engine stand - Prepare to stable engine stand that will prevent the engine from falling over, the set the engine assembly on the stand and secure it in position. - Drain engine oil from oil pan (the amount of oil differs according to the machine model) <u>Rotate the crankshaft engine 720° for check engine condition (jamed or not)</u> <table><tr><td>CHECK</td><td>YES</td><td></td></tr><tr><td></td><td>NO</td><td></td></tr></table>	CHECK	YES			NO									
CHECK	YES															
	NO															
2	Starter Motor Alternator	- Please note if not fitted to engine. - Please note if not fitted to engine. Starting Motor AssyS/N, UPPER : LOWER: ALTERNATOR ASSY S/N:				CHECK BY SUB ASSY 3 										
3	Turbocharger	- Remove Turbocharger, mounting bolts and adapters. - Remove turbo oil line fittings. Check pipes and connections. TURBOCHARGER ASSY S/N:				CHECK BY SUB ASSY 1 										
4	Exhaust Manifold	Remove and inspect for cracks, pulled threads, snapped studs and missing bolts.	<table><tr><td>FRONT</td><td></td><td></td></tr><tr><td>CENTRE</td><td></td><td></td></tr><tr><td>REAR</td><td></td><td></td></tr></table>	FRONT			CENTRE			REAR						
FRONT																
CENTRE																
REAR																
5	Air Intake Conector	Remove air intake conector assembly and check condition for cracked and damage.														
6	Water Pump Thermostat	- Remove tube and water pump assembly - Check for evidence of leakage. - Check for signs of wear and damage. - Remove Thermostat Assembly				CHECK BY SUB ASSY 1 										
7	Oil Cooler	- Check oil cooler housing for cracks and pulled threads. - Check oil cooler for dents and leaks.				CHECK BY SUB ASSY 2 										
8	Fuel Filter Ass'y	Remove tube and fuel filter assembly and check pulled threads, cracks and damage														

DISASSEMBLY CHECK SHEET		KOMATSU ENGINE 6D170E-5		JOB NUMBER		ENGINE MODEL		ENGINE SER. NO		MACHINE MODEL		DOCUMENT NO.		PUBL. DATE		REVISION NO.		PAGE	
						SAA6D170E-5								09/08/2010		01			
NO	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REUSE	SALVAGE	REPLACE	SKETCH DRAWING OR PHOTOES				REMARKS									
9	Air Intake Manifold	- Remove fuel inlet tube, timing rail tube, fuel rail tube and check for cracks and damage - Remove intake manifold assembly and check condition for cracks and damaged - Check for pulled threads and missing bolts. - Remove aftercooler and check for damage, water leaks, dust entry etc. - Check aftercooler water tubes for cracks, corrosion and any other damage.				CHECK BY SUB ASSY 2  													
10	ECM	damage, missing parts and wear. Remove Fuel Piping Fuel Pump S/N: Remove ECM P/N ASSY : S/N : ESN :																	
11	Oil Pan	- Remove oil Filler tube and check for cracked and damage - Remove and check oil pressure sensor - Remove pan and check for dents and cracks. - Remove and check drain plug for pulled threads - Remove brackets and then remove suction pipe - Check oil pick up pipe for cracks in the welds and check screen for damage				 													
12	Head Cover	Check for cracks and damage to seal face.				 													
13	Rocker Arm and Shaft	- Remove rocker arms and shafts and check for wear and other damage. - Check the contact area on the crossheads are from scuffing and wear. - Pull out the pushrods and check the pushrods for bends and other damage. - Check adjusting screw for cracks, damage and wear.				CHECK BY SUB ASSY 1  													
14	Rocker Arm Housing	- Remove rocker arm housing and check condition for cracked, pulled threads wear and damage. ★ Fit tags and keep in sets for each cylinder no.				 													
15	Injector Ass'y	- Remove holder mounting bolts and Using special tool to pull out of the injector assemblies - Check O-Ring contact area, Filter clogs and damage																	

DISASSEMBLY CHECK SHEET			JOB NUMBER		DOCUMENT NO.		RC/QR02/EG/01/03	
KOMATSU ENGINE 6D170E-5			ENGINE MODEL		PUBL. DATE		09/08/2010	
			ENGINE SER. NO		REVISION NO.		01	
			MACHINE MODEL		PAGE			
NO	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REUSE	SALVAGE	REPLACE	CHECK BY SUB ASSY 1	SKETCH DRAWING OR PHOTOS	REMARKS
16	Cylinder Head	- Remove Cylinder head assembly and check cylinder head bolts for pin punch marks. If there are five marks, renew bolts. - Check the exhaust face condition and bolt holes for pulled threads. - Check the head surface for water marks or damage. - Doing Colour Check inspection joint with QA Officer						
17	Cam Followers	- Inspect cam followers for damage to the contact surface of the rollers. - Check condition of covers.						
	Vibration Damper and Pulleys	- Remove and check vibration damper for damage. - Check pulleys for damage and wear. - Check belt tensioner for wear. - Check front trunnion mount.						
19	Flywheel	- Remove Flywheel and check bolts and bolt holes. - Check pilot bearing hole. - Inspect ring gear for chipped and worn teeth.						
20	Under plate	Remove underplate and check dents or damage						
21	Flywheel Housing	Remove Flywheel housing and check Flywheel housing for cracks, check bolt holes.						
22	Front Support and Front Cover	- Remove and inspect front support for damage - Check surface condition of seal running area on front shaft.						
23	Front Gear Ass'y	- Inspect and record condition of individual gears. Check for chipped teeth, cracks, excessive backlash and end play - Remove Gears - Check shafts and bearings for wear. - Measure thickness of thrust plates for wear.						
24	Oil Pump assembly	- Remove Oil Pump Ass'y and check pump - O-rings at inlet and outlet ports and check pump for wear and damage.						

DISASSEMBLY CHECK SHEET		JOB NUMBER		DOCUMENT NO.		RC/QR02/EG/01/03															
KOMATSU ENGINE 6D170E-5		ENGINE MODEL		PUBL. DATE		09/08/2010															
		ENGINE SER. NO		REVISION NO.		01															
		MACHINE MODEL		PAGE																	
NO	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REUSE	SALVAGE	REPLACE	SKETCH DRAWING OR PHOTOES	REMARKS														
25	Camshaft	- Check end float of camshaft. - Pull out Camshaft and check camshaft journals for wear. - Check cam surfaces for wear and other damage - Check camshaft for bends.				CHECK BY MACHINING 															
26	Rear Idler Gear Ass'y	- Check gears backlash and end play - Remove the idler gear assembly and check - Bushing and Shaft wear, Gear teeth chipping and wear				CHECK BY MACHINING 															
27	Piston Cooling Nozzles	Check for bends, cracks and blockages. grooves and skirts <table border="1"> <tr> <td>Cyl. No.</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table> Notes : ☒ Good ☐ No Good	Cyl. No.	1	2	3	4	5	6											CHECK BY SUB ASSY 1 	
Cyl. No.	1	2	3	4	5	6															
28	Connecting Rods	- Check con rod for cracks and damage. - Check big end bore for scratches or distortion. - Check for damage on mating face of cap. - Measured end play to connecting rod <table border="1"> <tr> <td>Cyl. No.</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table> STD : 0.200 ~ 0.374	Cyl. No.	1	2	3	4	5	6											CHECK BY MACHINING 	
Cyl. No.	1	2	3	4	5	6															
29	Pistons	Measure wear on piston skirt, ring grooves and pin bore <table border="1"> <tr> <td>Cyl. No.</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table> Notes : ☒ Good ☐ No Good	Cyl. No.	1	2	3	4	5	6												
Cyl. No.	1	2	3	4	5	6															
30	Crankshaft	- Inspect main and pin journals for scratches, wear, cracks and discoloration. - Check that oil holes are not blocked. - Check fillet area for scratches. - Determine wear by measuring ϕ of journals. - Check seal surface condition. - Punch Mark engine S/N rear surface of crankshaft (Marking on free area contact with flywheel) CRANKSHAFT S/N:				CHECK BY MACHINING 	MEASUREMENT RESULT 1. CRANKSHAFT ENDPLAY ENDPLAY:.....MM 2. FACE RUN OUT FRONT R/OUT:.....MM REAR R/OUT:.....MM 3. RADIAL RUN OUT FRONT RADIAL:.....MM REAR RADIAL:.....MM														
31	Main cap	- Check cap bolts for number of punch marks. - Check cap fit to block. - Punch S/N engine at the front face of all main bearing cap match with S/N engine block - Doing Colour Check inspection joint with QA Officer - Check trust metal kondition, cup and block must Flat. For detail see picture beside - thrust metal seat on no.6 man cup for engine 170-3/-5																			

DISASSEMBLY CHECK SHEET		JOB NUMBER		DOCUMENT NO.		RC/QR02/EG/01/03						
KOMATSU ENGINE 6D170E-5		ENGINE MODEL		SAA6D170E-5		PUBL. DATE						
		ENGINE SER. NO				REVISION NO.						
		MACHINE MODEL				PAGE						
NO	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED						REUSE	SALVAGE	REPLACE	SKETCH DRAWING OR PHOTOES	REMARKS
32	Cylinder Liners	- Check for pitting and cracks on the outer circumference of the Cylinder liner - Inspect the inner surface for scoring and corrosion. - Check for pitting damage on seal grooves. - Measure inside Ø, roundness and cylindricity of Cylinder liner										
	Cyl. No.	1	2	3	4	5	6					
	Notes : √ Good X No Good											
33	Cylinder Block	- Check for cracks in the block, especially in the bolt hole part of the cylinder head. - Inspect the lower liner seal area. - Check the blind plug surface for corrosion. - Check all threads. - Check mounting surface of cylinder head for distortion and corrosion. - Check for clogging of oil and water holes. - Doing Colour Check inspection joint with QA Officer									CHECK BY MACHINING 	

DISASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D170E-5

JOB NUMBER		DOCUMENT NO.	RC/QR02/EG/01/03
ENGINE MODEL	SAA6D170E-5	PUBL. DATE	09/08/2010
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MACHINE MODEL		PAGE	

FRONT SIDE TIMING GEAR MEASUREMENT

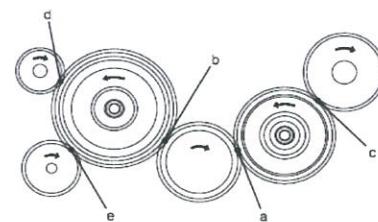
Position	Backlash (mm)	Measured	Check
A	Crankshaft gear and Idle gear (Medium) 0.114 ~ 0.320 (R/L: 0.5)		
B	Crankshaft gear and Idle gear (large) 0.134 ~ 0.362 (R/L:)		
C	Idle gear (Medium) and Fuel pump drive gear 0.114 ~ 0.320 (R/L: 0.5)		
D	Idle gear (Large) and Water pump drive gear 0.121 ~ 0.333 (R/L: 0.5)		
E	Idle gear (Large) and Oil pump drive gear 0.121 ~ 0.333 (R/L: 0.5)		

No.	End play (mm)	Measured	Check
1	Oil pump drive gear 0.04 ~ 0.18 (C/L : 0.34)		
2	Water pump drive gear 0.04 ~ 0.20 (C/L : 0.34)		
3	Idler gear 0.04 ~ 0.17 (C/L : 0.30)		
4	Crankshaft gear 0.04 ~ 0.17 (C/L : 0.30)		
5	Idle gear 0.04 ~ 0.17 (C/L : 0.34)		
6	Alternator fuel pump drive gear 0.04 ~ 0.17 (C/L : 0.34)		

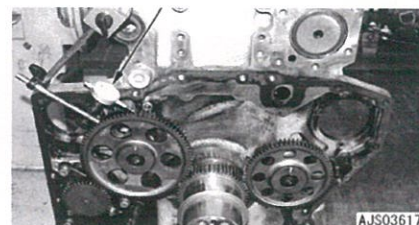
REAR SIDE TIMING GEAR MEASUREMENT

Position	Backlash (mm)	Measured	Check
A	Crankshaft gear and Idle gear (large) 0.155 ~ 0.412 (R/L: 0.6)		
B	Crankshaft gear and Idle gear (Medium) 0.145 ~ 0.380 (R/L: 0.6)		

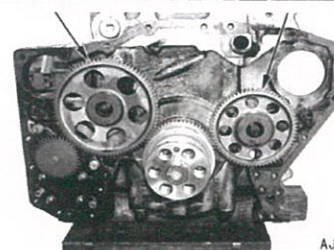
No.	End play (mm)	Measured	Check
1	Camshaft drive gear 0.05 ~ 0.20 (C/L : 0.30)		
2	Idle gear 0.04 ~ 0.17 (C/L : 0.34)		
3	Crankshaft drive gear 0.04 ~ 0.17 (C/L : 0.30)		



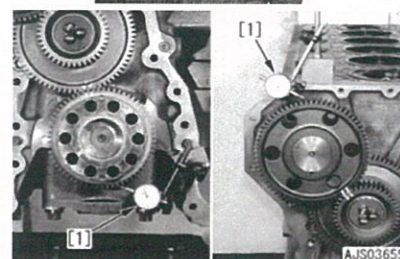
CWE00701



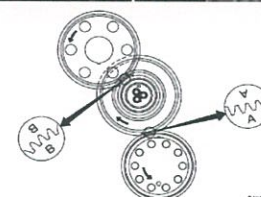
AJS03617



AJS03605



AJS03655



CWE00665

REMARKS :

INSPECTION BY

APPROVAL BY

(MECHANIC)

GL/SPV PRODUCTION

DISASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D170E-5

JOB NUMBER		DOCUMENT NO.	RC/QR02/EG/01/03
ENGINE MODEL	SAA6D170E-5	PUBL. DATE	09/08/2010
ENGINE SER. NO		REVISION NO.	01
MACHINE MODEL		PAGE	

ATTACHED
OUTSIDE DIAMETER OF CRANKSHAFT PIN JOURNAL AND MAIN JOURNAL

6D95 Series

Unit : mm

Size	Main Journal		Pin Journal	
	Standard Size	Repair Limit	Standard Size	Repair Limit
Standard	70,00	69,86	57,00	56,91
0.25 US	69,75	69,91	56,75	56,66
0.50 US	69,50	69,36	56,50	56,41
0.75 US	69,25	69,11	56,25	56,16
1.00 US	69,00	68,86	56,00	55,91
Out of Roundness	0 ~ 0.010	0,02	0 ~ 0.010	0,02
Tolerance : - 0.015				

6D125 Series

Unit : mm

Size	Main Journal		Pin Journal	
	Standard Size	Repair Limit	Standard Size	Repair Limit
Standard	110,00	109,98	80,00	79,88
0.25 US	109,75	109,63	79,75	79,63
0.50 US	109,50	109,63	79,50	79,38
0.75 US	109,25	109,38	79,25	79,13
1.00 US	109,00	109,13	79,00	78,88
Out of Roundness	0 ~ 0.010	0,02	0 ~ 0.010	0,02
Tolerance : - 0.015 ~ - 0.070				

6D170 Series

Size	Main Journal		Pin Journal	
	Standard Size	Repair Limit	Standard Size	Repair Limit
Standard	140,00	139,91	108,00	107,91
0.25 US	139,75	139,66	107,75	107,66
0.50 US	139,50	139,41	107,50	107,41
0.75 US	139,25	139,16	107,25	107,16
1.00 US	139,00	139,91	107,00	106,91
Out of Roundness	0 ~ 0.010	0,01	0 ~ 0.010	0,01
Tolerance : - 0.025				

6D140 Series

Unit : mm

Size	Main Journal		Pin Journal	
	Standard Size	Repair Limit	Standard Size	Repair Limit
Standard	120,00	119,83	90,00	89,88
0.25 US	119,75	119,63	89,75	89,63
0.50 US	119,50	119,83	89,50	89,38
0.75 US	119,25	119,13	89,25	89,13
1.00 US	119,00	118,88	89,00	88,88
Out of Roundness	0 ~ 0.010	0,02	0 ~ 0.010	0,02
Tolerance : - 0.020				

6D155 Series

Unit : mm

Size	Main Journal		Pin Journal	
	Standard Size	Repair Limit	Standard Size	Repair Limit
Standard	125,00	124,70	102,00	101,75
0.25 US	124,75	124,45	101,75	101,50
0.50 US	124,50	124,20	101,50	101,25
0.75 US	124,25	123,95	101,25	101,00
1.00 US	124,00	123,70	101,00	100,75
Out of Roundness	0.010 max	0,02	0.010 max	0,02
Tolerance	- 0.060 ~ (-0.080) (Main Journal)			
	- 0.050 ~ (-0.080) (Pin Journal)			

12V140 Series

Unit : mm

Size	Main Journal		Pin Journal	
	Standard Size	Repair Limit	Standard Size	Repair Limit
Standard	148,00	147,98	100,00	99,88
0.25 US	147,75	147,73	99,75	99,63
0.50 US	147,50	147,48	99,50	99,38
0.75 US	147,25	147,23	99,25	99,13
1.00 US	147,00	146,98	99,00	98,88
Out of Roundness	0 ~ 0.010	0,01	0 ~ 0.010	0,01
Tolerance : - 0.020				

DISASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D170E-5

JOB NUMBER		DOCUMENT NO.	RC/QR02/EG/01/03
ENGINE MODEL	SAA6D170E-5	PUBL. DATE	09/08/2010
ENGINE SER. NO		REVISION NO.	01
MACHINE MODEL		PAGE	

OUTSIDE DIAMETER OF CAMSHAFT JOURNAL AND CAM HEIGHT

6D95 Series Unit : mm

Cam Height			
Intake	42.7 ± 0.1	RL	42,20
Exhaust	43.4 ± 0.1	RL	42,90
OD Chamshaft Journal			
STD	- 0.080		
	50,50 - 0.110		

6D125 Series Unit : mm

Cam Height			
Intake	+ 0.395 52 + 0.195	RL	51,73
Exhaust	+ 0.395 52 + 0.195	RL	51,73
OD Chamshaft Journal			
STD	- 0.080		
	60 - 0.110		

6D170 Series Unit : mm

Cam Height			
Intake	+ 0.4144 62 + 0.2144	RL	61,37
Exhaust	+ 0.3215 61 + 0.5215	RL	59,64
OD Chamshaft Journal			
STD	- 0.080		
	72 - 0.110		

6D140 Series Unit : mm

Cam Height			
Intake	55.48 ± 0.1	RL	55,08
Exhaust	55.75 ± 0.1	RL	55,35
OD Chamshaft Journal			
STD	- 0.016		
	65 - 0.036		

6D155 Series Unit : mm

Cam Height			
Intake	+ 0.355 51 + 0.345	RL	51,73
Exhaust	+ 0.355 51 + 0.345	RL	51,73
OD Chamshaft Journal			
STD	+ 0.080		
	61 + 0.110		

12V140 Series Unit : mm

Cam Height			
Intake	55.48 ± 0.1	RL	55,08
Exhaust	55.75 ± 0.1	RL	55,35
OD Chamshaft Journal			
STD	- 0.016		
	65 - 0.036		



PISTON PIN

MEASURING AND POLISHING

INSPECTION CHECK SHEET

KOMATSU ENGINE 6D170-5 Series

MACHINE MODEL

JOB NO:

ENGINE ASSY SECTION

PISTON PIN MEASURING AND POLISHING CHECK SHEET

WO NUMBER		PART NAME	PISTON PIN	DOCUMENT NO.	RC/QR03/EG/05/03
ENGINE MODEL	6D140-5	PART NUMBER	6248-31-2410	PUBL. DATE	
ENGINE S/N		START DATE / TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 1

MACHINIST NAME	
----------------	--

MAN HOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Measuring			
TOTAL			

STANDARD (unit mm)	Clearance limit
6D170E-3 = Ø 67.980 - 67.974	0,11
6D170E-2 = Ø 68.000 - 67.940	0,11
6D170E-5 = Ø 68.000 - 67.994	0,11

QUANTITY	
----------	--

INSPECTION MEASURING PROCESS

CYL.NO.	CHECK POSITION	1			2			3			Clearance actual
		MAC	QA	SPV	MAC	QA	SPV	MAC	QA	SPV	
1	X - X'										
	Y - Y'										
2	X - X'										
	Y - Y'										
3	X - X'										
	Y - Y'										
4	X - X'										
	Y - Y'										
5	X - X'										
	Y - Y'										
6	X - X'										
	Y - Y'										

MECHANIC INSPECTION COMMENT :

INSPECTION BY :

INSPECTION & APPROVED BY :

QR06-SAL-005-03

REV. 04



PT.SAPTA INDRA SEJATI

REBUILD CENTER

**CAMSHAFT ASSY
MEASUREMENT PROCESS AND
INSPECTION CHECK SHEET
KOMATSU ENGINE 6D170E-5**

MACHINE MODEL

WO NO. :

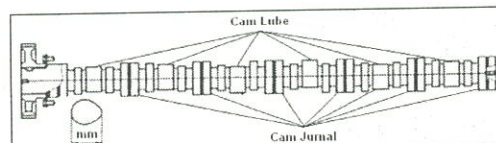
ENGINE SECTION

CAMSHAFT PROCESS AND MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CAMSHAFT ASSY	DOCUMENT NO.	
ENGINE MODEL	140,170,12V140 SERIES	PART NUMBER		PUBL. DATE	16/04/2010
ENGINE S/N		START DATE/TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

MECHANIC NAME	
---------------	--

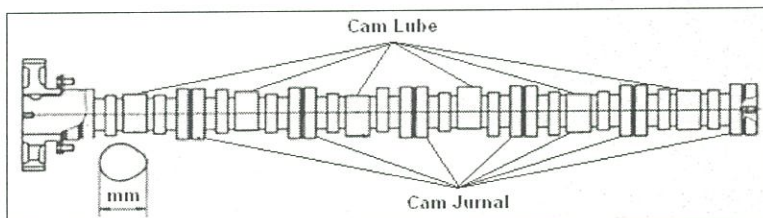
MAN HOURS UTILIZATION			
	START	FINISH	MAN HOURS
Machining Process			
Measuring Process			
TOTAL			



Unit : mm

STANDARD		REPAIR LIMIT	No	CAMSHAFT JOURNAL		CAM LUBE INTAKE		CAM LUBE EXHAUST	
				MEC	GL/SVP/QA	MEC	GL/SVP/QA	MEC	GL/SVP/QA
12V140	Intake = 55.58 ~ 55.38	55,08	1	X					
	Exhaust = 55.65 ~ 55.85	55,35		Y					
	Cam Journal = 64.984 ~ 64.964		2	X					
				Y					
6D140	Intake = 55.58 ~ 55.38	55,08	3	X					
	Exhaust = 55.85 ~ 55.65	55,35		Y					
	Cam Journal = 64.984 ~ 64.964		4	X					
				Y					
6D170-3	Intake = 89.908 ~ 89.708	89,20	5	X					
	Exhaust = 90.268 ~ 90.068	88,80		Y					
	Cam Journal = 104.958 ~ 104.928	104,88	6	X					
				Y					
6D170-5	Intake = 78.065 ~ 78.336	78,00	7	X					
	Exhaust = 77.074 ~ 77.355	77,00		Y					
	Cam Journal = 89.987 ~ 90.035	89,98							

DESCRIPTION	OK	Not OK
CRACK HAS BEEN CHECK		



MECHANIC INSPECTION COMMENT :

INSPECTION BY :

INSPECTION & APPROVED BY :

MACHINIST (MACH)

QUALITY ASSURANCE (QA)

PROD.SUPERVISOR (SPV)

QR06-SAL-005-03

REV. 04



PT.SAPTA INDRA SEJATI

REBUILD CENTER

**CRANKSHAFT ASSY
MEASUREMENT PROCESS AND
INSPECTION CHECK SHEET**

(BEFORE SALVAGING)

KOMATSU ENGINE 6D170E-5

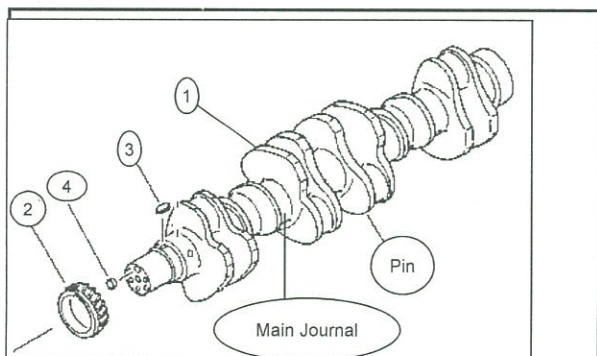
MACHINE MODEL

WO NO. :

ENGINE SECTION

CRANK SHAFT DISASSEMBLY INSPECTION AND MEASUREMENT CHECKSHEET

WO NUMBER		PART NAME	CRANKSHAFT	DOCUMENT NO.	QR08-SAL-011-03
ENGINE MODEL	6D170E-3/-5	PART NUMBER		PUBL. DATE	01/04/2009
ENGINE S/N		START DATE / TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 1



Physical Data

No.	Part Number	Part Name	Qty	Good	No Good
1		Crankshaft	1		
2		Gear	1		
3		Key	1		
4		Dowel Pin	1		

Standard size					Unit : mm
SIZE	MAIN JOURNAL		PIN JOURNAL		
	STD SIZE	REPAIR LIMIT	STD SIZE	REPAIR LIMIT	
STANDARD	140.000 ~ 139.975	139.91	108.000 ~ 107.978	107.91	
Under Size 0.25	139.750 ~ 139.725	139.66	107.750 ~ 107.728	107.66	
Under Size 0.50	139.500 ~ 139.475	139.41	107.500 ~ 107.478	107.41	
ROUNDNESS	0 ~ 0.010	0 ~ 0.015	0 ~ 0.010	0 ~ 0.015	
FILLET RADIUS	6.0 ~ 6.5	-	6.0 ~ 6.5	-	

MANHOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Washing / Cleaning			
Measuring Process			
TOTAL			

VISUAL CHECK	OK	Not OK
Crack, Scratch, Wear Has been checked		

FILLET RADIUS	Main Journal	Pin Journal

INSPECTION AND MEASUREMENT BY QA & MECHANIC DIS-ASSEMBLY

CRITICAL INSPECTION		RANDOM INSPECTION		ROUNDNESS	CRITICAL INSPECTION FILLET RADIUS		RANDOM INSPECTION		REMARK
		CHECK POINT					CHECK POINT		
		X - X	Y - Y				ACTUAL MEASUREMENT		
MAIN JOURNAL	1				MAIN JOURNAL	1			
	2					2			
	3					3			
	4					4			
	5					5			
	6					6			
	7					7			
PIN JOURNAL	1				PIN JOURNAL	1			
	2					2			
	3					3			
	4					4			
	5					5			
	6					6			

Main Journal & Pin Journal

Actual : STD or US

Crank Shaft Condition : Good or No Good

Inspection result :

MEASURING BY DIS ASSY	
MEASURING DATE	:
MECHANIC NAME	:
SIGN	:

RANDOM INSPECTION BY QA	
MEASURING DATE	:
INSPECTOR NAME	:
SIGN	:

QR06-SAL-005-03

REV. 04



PT.SAPTA INDRA SEJATI

REBUILD CENTER

**CONNECTING ROD
MEASUREMENT PROCESS AND
INSPECTION CHECK SHEET
KOMATSU ENGINE 6D170E-5**

MACHINE MODEL

WO NO. :

ENGINE SECTION

CONNECTING ROD SALVAGING PROCESS AND INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CONNECTING ROD ASSY	EKS UNIT CODE	
ENGINE MODEL	6D170E - 5	PART NUMBER		PUBL. DATE	16/04/2010
ENGINE S/N		START DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		DOCUMENT NO.	2 of 3

SMALL END AFTER SALVAGE PROCESS (NEW BEARING)						
CON-ROD NO.	TIGHTENING BOLTS	CHECK POINT	Ø SMALL END WITHOUT BEARING			
			X - X'		Y - Y'	
			MAC	GL/SVP/QA	MAC	GL/SVP/QA
1		A				
		B				
2		A				
		B				
3		A				
		B				
4		A				
		B				
5		A				
		B				
6		A				
		B				

CON ROD BENDING				
CON-ROD NO.	BEND		TWIST	
	Indicator Reading	Actual Bending	Indicator Reading	Actual Twist
1				
2				
3				
4				
5				
6				

BEND AND TWIST
Measure with formula :

Distance between hole

Width seat metal

x

1 Reading

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVAL BY :

MECHANIC (MEC)

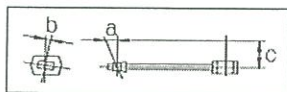
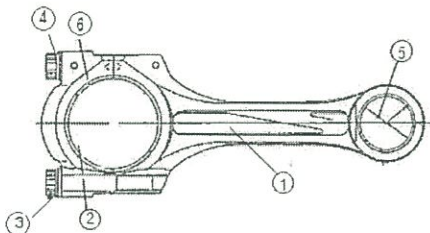
QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)

CONNECTING ROD SALVAGING PROCESS AND INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CONNECTING ROD ASSY	EKS UNIT CODE	
ENGINE MODEL	6D170E - 5	PART NUMBER		PUBL. DATE	16/04/2010
ENGINE S/N		START DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		DOCUMENT NO.	of 3

SKETCH DRAWING



Physical Data

MECHANIC NAME				
Standard size				
Unit : mm				
ENGINE MODEL	TIGHTENING BOLT STEP PROCESS WITHOUT BEARING (METAL) KGM + ANGLE / °	DIAMETER SMALL END WITHOUT BEARING	DIAMETER SMALL END WITH BEARING	REPAIR LIMIT
6D140 - 3	12.5 ~ 13.5 + 90 ° ~ 120 °	57.400 ~ 57.430	52.030 ~ 52.049	52.09
6D170 - 3	19.0 ~ 21.0 + 90 ° ~ 105 °	75.000 ~ 75.030	68.010 ~ 68.049	-
6D170 - 5	19.0 ~ 21.0 + 90 ° ~ 105 °	75.000 ~ 75.030	68.030 ~ 68.049	-
ENGINE MODEL	DIAMETER BIG END WITHOUT BEARING	BEND (a)	TWIST (b)	DISTANCE BETWEEN HOLE
6D140 - 3	94.992 ~ 95.042	RL 0.25	RL 0.35	264
6D170 - 3	115.000 ~ 115.026	RL 0.10	RL 0.25	305
6D170 - 5	115.000 ~ 115.026	RL 0.10	RL 0.25	305

No.	Part Number	Part Name	Qty	New	Reuse	VISUAL CHECK		
1		Connecting rod				CRACK HAS BEEN CHECKED		
2		Connecting rod cap						
3		Connecting rod bolt				MANHOURS UTILIZATION		
4		Washer				START	FINISH	MAN HOURS
5		Crank pin Metal				Measuring Process		
						TOTAL		

BIG END AFTER SALVAGING PROCESS (NEW BEARING)

CON-ROD NO.	TIGHTENING BOLTS	CHECK POINT	Ø BIG END WITHOUT BEARING					
			X - X'		Y - Y'		Z - Z'	
			MEC	GL/SVP/QA	MEC	GL/SVP/QA	MEC	GL/SVP/QA
1		A						
		B						
2		A						
		B						
3		A						
		B						
4		A						
		B						
5		A						
		B						
6		A						
		B						

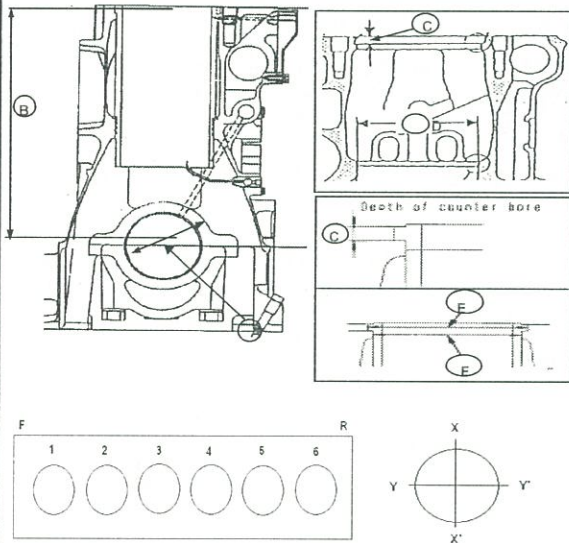


PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

CYLINDER BLOCK MEASURING PROCESS AND INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CYLINDER BLOCK	EKS UNIT CODE	
ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	10/04/2010
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE / TIME		DOCUMENT NO.	bf 3

SKETCH DRAWING



MECHANIC NAME

Cyl. Block Inspection

Standard Size

Unit : mm

DESCRIPTION	STANDARD SIZE
A MAIN BEARING BORE	147.999 ~ 148.025
B CYLINDER BLOCK HEIGHT	488.960 ~ 489.040
C USE OVER SIZE GASKET	488.560 ~ 488.960
D COUNTER BORE DEPTH	14.000 ~ 14.050
E CYLINDER LINER O-RING LAND	190.340 ~ 190.400
F CYLINDER LINER FLANGE SEAT	205.965 ~ 206.015
F COUNTER BORE DIAMETER	194.565 ~ 194.615

TIGHTENING TORQUE OF MAIN BEARING CAP

1 st step	=	27.5 ~ 30.5 Kgm
2 nd step	=	57.0 ~ 59.0
3 rd step	=	90° ~ 120°

VISUAL CHECK

OK

Not OK

CRACK HAS BEEN CHECKED

MANHOURS UTILIZATION

	START	FINISH	MAN HOURS
Measuring Process			
TOTAL			

INSPECTION SALVAGE PROCESS BY MECHANIC

RESULT : ☒ GOOD CONDITION

X : BAD CONDITION

	Main Bore No.	X - X'		Y - Y'	
		A	B	A	B
A	1	A			
		B			
	2	A			
		B			
	3	A			
		B			
	4	A			
		B			
	5	A			
		B			
	6	A			
		B			
	7	A			
		B			

CYLINDER LINER O-RING LAND VISUAL CHECK

	Cyl.	No.	CHECK			Cyl.	No.	CHECK	
			MECH	GL/QA/SVP				MECH	GL/QA/SVP
D		1					4		
		2					5		
		3					6		



PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

CYLINDER BLOCK MEASURING PROCESS AND INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CYLINDER BLOCK	EKS UNIT CODE	
ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	10/04/2010
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE / TIME		DOCUMENT NO.	bf 3

CYLINDER LINER FLANGE SEAT						
	E	CYLINDER LINER FLANGE SEAT				
	F	COUNTER BORE DIAMETER				
<table border="1" style="width: 100%;"> <tr> <td style="width: 30%;"></td> <td style="width: 30%; text-align: center;">205.965 ~ 206.015</td> </tr> <tr> <td style="width: 30%;"></td> <td style="width: 30%; text-align: center;">194.565 ~ 194.615</td> </tr> </table>				205.965 ~ 206.015		194.565 ~ 194.615
	205.965 ~ 206.015					
	194.565 ~ 194.615					
Note : 						

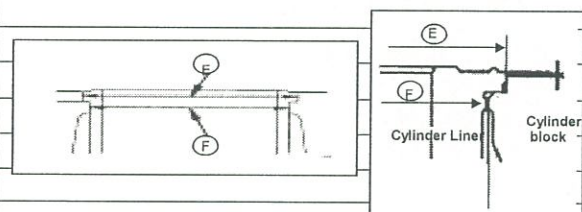
INSPECTION PROCESS BY MECHANIC

CYLINDER LINER FLANGE SEAT										
	Cyl	X - X'					Y - Y'			
		CHECK					CHECK			
		MEC	GL/QA				MEC	GL/QA		
E	1					1				
	2					2				
	3					3				
	4					4				
	5					5				
	6					6				

INSPECTION PROCESS BY MECHANIC

COUNTER BORE DIAMETER										
	Cyl	X - X'					Y - Y'			
		CHECK					CHECK			
		MEC	GL/QA				MEC	GL/QA		
F	1					1				
	2					2				
	3					3				
	4					4				
	5					5				
	6					6				

INSPECTION RESULT :



INSPECTION BY :

INSPECTION & APPROVAL BY

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)



PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

CYLINDER BLOCK MEASURING PROCESS AND INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CYLINDER BLOCK	EKS UNIT CODE	
ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	10/04/2010
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE / TIME		DOCUMENT NO.	bf 3

INSPECTION PROCESS

COUNTER BORE DEPHT

Cyl	AFTER REPAIR											
	X			X'			Y			Y'		
	MECH	QA		MECH	QA		MECH	QA		MECH	QA	
1												
2												
3												
4												
5												
6												

INSPECTION RESULT :

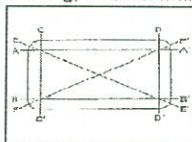
Roughness surface counter bore		MECH	QA	No.	Place	Contents	Reuse Judgement Standard	Method of Judgement	Measurement Instrument	Remanufacturing Method	Remanufacturing Standard
	A			1	Upper surface	Scratches	Depth under 0.1mm	Visual Inspection		Polish with Oil stone	No convex portion under 0.1mm
	B				Sealing & Grommet of Gasket	Rust	Depth under 0.2mm	Visual Inspection		Machining oversize	Depth under 0.3mm
	C							Feel fingernail			
	D					Wear	Surface roughness under 12.5s	Measurement	Roughness checker	Grinding surface	Roughness under 12.5s
	E										
	F										

Flatness of Top Surface of Cylinder block		MECH	QA									
A												
B												
C												
D												
E												
F												

Measurement procedure of Flatness Cylinder block surface

Inspect the head gasket mount surface on the top deck surface for fretting, corrosion and scratches.

Depth of fretting, corrosion or scratches	Remedy
Less than 0.1mm	Recondition with oil stone.
Less than 0.4mm	Resurface the top deck
More than 0.4mm	Replace the block



INSPECTION BY :

INSPECTION & APPROVAL BY

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)

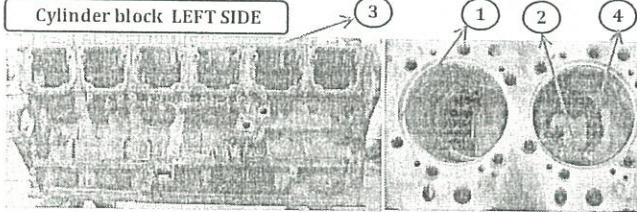
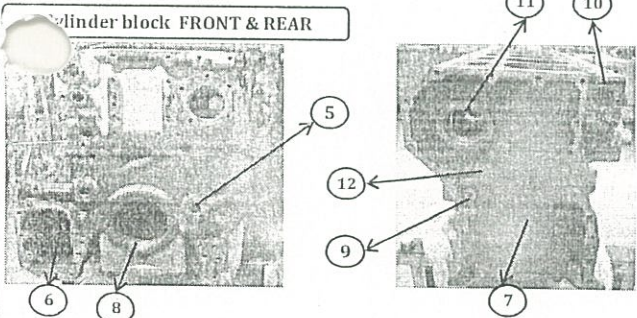
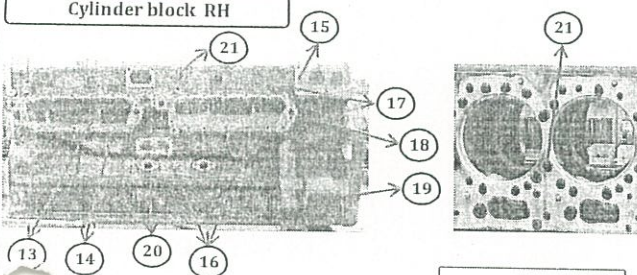
CYLINDER BLOCK INSPECTION SHEET ENGINE

MACHINE MODEL		CYLINDER BLOCK P/N	6245-21-1100	DOCUMENT NO.	ENG-CB-6D170/3-2-001
ENGINE MODEL	SA/SAA6D170-3,-5	INSPECTION DATE		PUBL. DATE	3-Mar-16
ENGINE NO.		WO NUMBER		REVISION NO.	03
CUSTOMER NAME				PAGE	2 OF 3

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO	ITEM CHECK	INSPECTION		REMARK
			OK	NOT OK	
Cylinder block LEFT SIDE 	1	Counter bore surface area			
	2	Water jacket wall area			
	3	Cylinder block upper surface			
	4	O ring liner area			
 Contoh pengecekan thread pada cyl block dengan bolt		Inspeksi pada semua thread, pengecekan dengan menggunakan bolt yang sesuai spec dengan thread yang ada di cyl block.			TOTAL THREAD 72 THREAD AND 7 PLUG
		# All thread on camfollower side			
		# All thread cyl head&Top surface LH cyl block			
		# All thread on (common rail side cyl block)			
	5	Timing gear hole area			
	6	Oil pump contact area			
	7	Thrust bearing surface			
	8	Main bore front surface			
	9	Oil and water hole gallery surface area			
	10	Blind cover water area			
	11	Camshaft hole surface rear area RH LH			
	12	Remove all plug on oil and water gallery			
Cylinder block FRONT & REAR 		All Thread condition			TOTAL THREAD 48 FRONT SIDE, 40 THREAD REAR SIDE
		Inspeksi pada semua thread, pengecekan dengan menggunakan bolt yang sesuai spec dengan thread yang ada di cyl block.			
		# All thread timing gear shaft & Camshaft			
		# All thread front cover mounting			
	13	Oil cooler hole area			
	14	Oil cooler mounting thread			
	15	Thermostat area			
	16	Water hole area			
	17	Spare thread on upper il cooler			
	18	Oil line oil cooler area			
	19	Big plug oil thread			
	20	Turbocharger drain hole thread			
Cylinder block RH 	21	Check all plug still complete			TOTAL THREAD 112 THREAD
		Inspeksi pada semua thread, pengecekan dengan menggunakan bolt yang sesuai spec dengan thread yang ada di cyl block.			
		# All thread Camfollower & Oil cooler area			
		# All thread oil adapter and under oil cooler			
	21.1	# All thread cyl head & all thread on surface			
	22	Piston cooling thread			
	23	Main cap mounting thread			
	24	All oil pan mounting thread			
	25	All Main metal upper seat area			
		Inspeksi pada semua thread, pengecekan dengan menggunakan bolt yang sesuai spec dengan thread yang ada di cyl block.			TOTAL THREAD UPPER 26 THREAD, LOWER 72 THREAD

Notes :

1. Semua thread yang tertutup dengan tanah (soil) harus dilakukan retap dan cleaning oleh man power disassy.

Checked by,

Inspected by,

Approved by,

(MECHANIC)

(QA INSPECTOR)

(PROD.SUPERVISOR)



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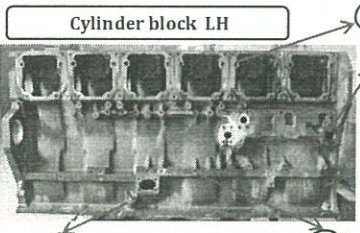
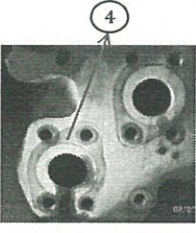
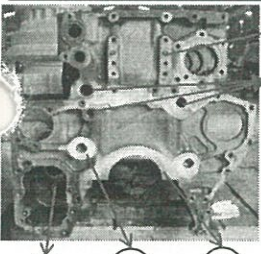
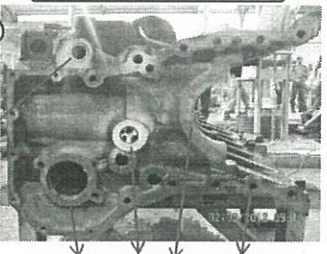
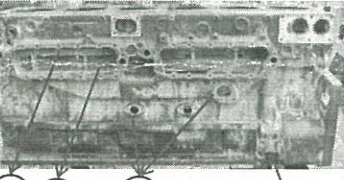
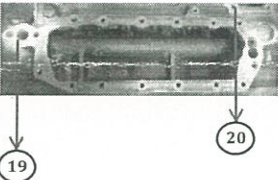
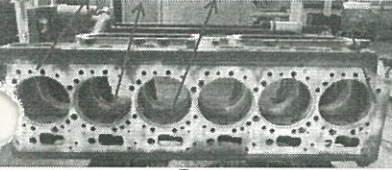
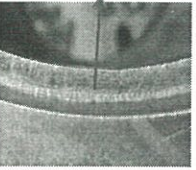
CYLINDER BLOCK INSPECTION SHEET ENGINE

MACHINE MODEL		CYLINDER BLOCK P/N	6240-21-1100	DOCUMENT NO.	ENG-CB-170/5-2-001
ENGINE MODEL	SAA6D170E-3	INSPECTION DATE		PUBL. DATE	6-Aug-14
ENGINE NO.		WO NUMBER		REVISION NO.	00
CUSTOMER NAME				PAGE	1 OF 1

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOS	NO	ITEM CHECK	COLOUR CHECK		REMARK
			YES	NO	
Cylinder block LH   1 2 3 4 5	1	Thread of camfollower mounting shaft			
	2	Thread of Plug oil adapter (Big Plug)			
	3	Thread of Oil level gauge bushing			
	4	Hole of cylinder block oil adapter			
	5	Check all thread condition on LH side			
Cylinder block FRONT   6 7 8 9 10 11	6	Timing gear front hole area (2 hole)			
	7	Main gallery and piston cooling gallery			
	8	Main metal seat surface (front engine)			
	9	Oil hole of oil pump contact surface			
	10	All thread condition in front side			
	11	Main gallery and piston cooling gallery			
	12	Camshaft thrust plate contact area			
	13	Timing gear rear hole			
	14	Main metal rear area			
	15	All thread condition			
Cylinder block RH     16 17 18 19 20 21 22 23 24 25 26 27	16	Oil cooler thread mounting			
	17	Oil cooler element area			
	18	Oil drain turbo and water adapter hole			
	19	Oil cooler hole front and rear (4 pcs)			
	20	Check all thread on RH side			
	21	Upper surface area			
	22	O ring liner area and chamfer condition			
	23	Water jacket area			
	24	Counter bore area			
	25	Counter bore condition from pitting and chipping			
	26	Remove all Oil and water gallery plug			
	27	Check RIB condition (reinforcing), make sure cyl block was PSN			

Notes :

Checked by,

Inspected by,

Approved by,

(MECHANIC)

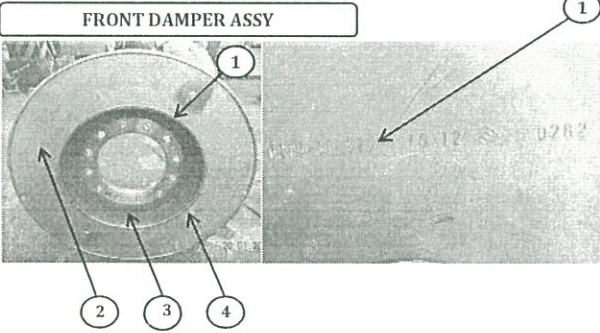
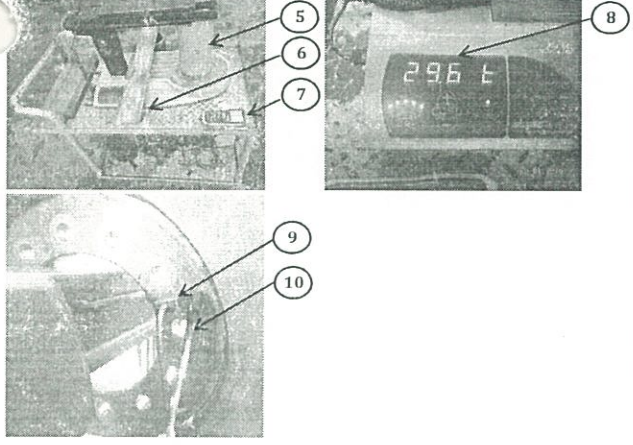
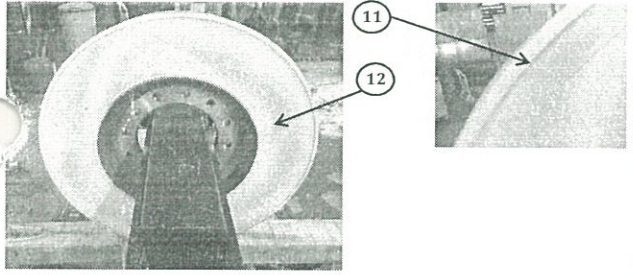
(QA INSPECTOR)

(PROD.SUPERVISOR)

FRONT DAMPER INSPECTION CHECKSHEET ENGINE

MACHINE MODEL		FRONT DAMPER P/N		DOCUMENT NO.	ENG-FD-6D170/3-2-001
ENGINE MODEL	6D170 SERIES	INSPECTION DATE		PUBL. DATE	21-Jan-16
ENGINE SERIAL NO.		WO NUMBER		REVISION NO.	00
CUSTOMER NAME				PAGE	1 OF 1

✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO	ITEM CHECK	INSPECTION		REMARK
			OK	NOT OK	
FRONT DAMPER ASSY 	1	Check Damper assy is spec for engine 6D170 series.			
	2	Check front damper assy from bend, chipped, pitted and dirty			
	3	Washing damper use washing chemical for remove oil from damper body.			
	4	Remove painting from damper body, use special paint remover.			
PREPARATION HEATER TOOL 	5	Prepare tool assy (bearing heater) for check damper condition			
	6	Setting block for damper stand and use beam on upper block.			
	7	setting Termometer for check heat temperatur when test damper. Maximum temperatur when test is 90° celcius.			
	8	Setting time on heater machine, time for test damper is 30 minutes.			
	9	Put heat sensor machine to damper body. Area contact is mounting area nor damper area.			
	10	Put the temperatur sensor of thermometer tool on damper mounting area.			
TEST DAMPER WITH HEATER 	11	After finish test, check with developer on damper contact cover (front side/part number side) for check abnormal leakage.			
	12	Before release the damper or finally check, make sure no leakage on this contact cover.			
	13	Remove damper and clean developer, and then delivery damper to assembly line.			

Notes :

Checked by,

Inspected by,

Aproved by,



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PLANT REBUILD CENTER

CRANKSHAFT INSPECTION SHEET ENGINE

MACHINE MODEL		CRANKSHAFT P/N	6240-21-1100	DOCUMENT NO.	ENG-CS-170/3-2-001
ENGINE MODEL	SA/SAA6D170E-3	INSPECTION DATE		PUBL. DATE	13-Feb-15
ENGINE NO.		WO NUMBER		REVISION NO.	00
CUSTOMER NAME				PAGE	1 OF 1

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO	ITEM CHECK	INSPECTION		REMARK
			GOOD	NO GOOD	
Crankshfat assy	1	Fillet radius Main and Pin journal no. 1 from pitting & Corrosion			
	2	Fillet radius Main and Pin journal no. 2 front and rear side from pitting &			
	3	Fillet radius Main and Pin journal no. 3 front and rear side from pitting and corrosion			
	4	Fillet radius Main and Pin journal no.4 from pitting and corrosion			
	5	Fillet radius Main and Pin journal no.5 from pitting and corrosion			
	6	Fillet radius Main and Pin journal no.6 from pitting and corrosion			
	7	Fillet radius main journal no.7 from pitting and corrosion			
	8	Check condition front & rear seal contact area from wear and fretting			
	9	Check crankgear condition from piiting and chipping. (front gear for engine 12V140 and front-rear gear for engine 170)			
	10	Check condition web,counter weight from chipping and pitting.			
	11	Check condition all thread crankshaft front and rear. Make sure engine S/N was marking on rear surface area			
Note: Engine serial Number position marking code,location on Rear Pilot bearing area (free contact area with Flywheel).					
Notes :					

Checked by,

Inspected by,

Aproved by,

(MECHANIC)

(QA INSPECTOR)

(DEPT. SUPERVISOR)

REV.00



PLANT REBUILD CENTER

CYL BLOCK

ASSEMBLY

CHECK SHEET

KOMATSU ENGINE 6D170-5

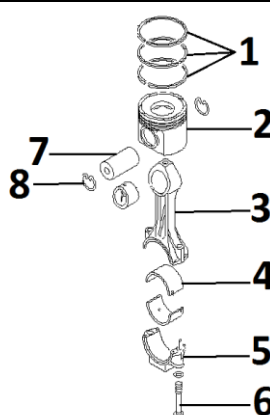
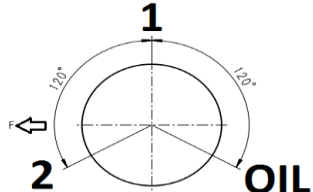
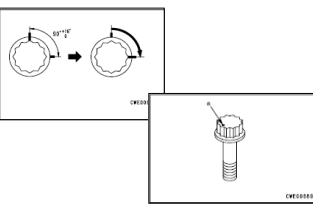
MACHINE MODEL

WO NO:

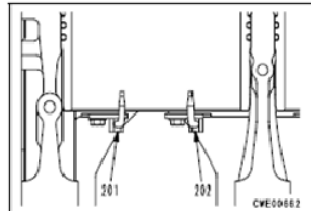
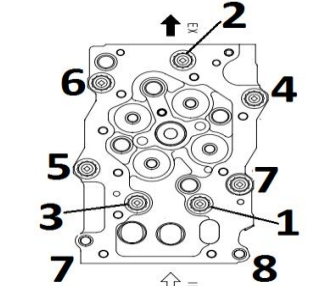
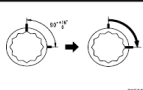
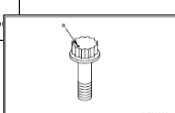
SHORTBLOCK SECTION

ASSEMBLY CHECKSHEET ENGINE ASSY KOMATSU ENGINE 6D170E-5			WO NUMBER		DOCUMENT NO	RC/QR04/EG/02/03													
			ENGINE MODEL		PUBLIC.DATE														
			ENGINE SERIAL NO.		REVISION NO.	08													
			MACHINE MODEL		PAGE														
PROCESS		START/date&time			ASSEMBLY BY	SHORT BLOCK													
		FINISH/date&time				MAIN LINE													
Firing Order			1 - 5 - 3 - 6 - 2 - 4			All dimensions in millimetres													
Valve clearance			Intake			0.32 ± 0.02 mm													
			Exhaust			0.62 ± 0.02 mm													
						All torques in Kilogram metres													
NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	ACTION DATE	SIGN	SKETCH DRAWING AND PHOTOS												
						MEC	GL/QA												
1	Cylinder block	Bores	Check that the liner counterbore is clean and																
		Inside of oil holes	Check that all oil holes are free from foreign matter such as chips or dirt. matter such as chips or dirt.																
		U-plugs & blind plugs	Make sure that all U-plugs and blind plugs have been installed																
			Make sure that cylinder block condition is ready to used completely.																
			Tightening Plug Hexagon plug 5 mm Act : kgm																
			Oversize head gasket required : YES NO																
	Cylinder Block S/N (NEW): Cyl.Block S/N (ORIGINAL):																		
	Engine S/N • Marking code OS on LH rear surface of cyl block(for cyl block after cutting surface only) <u>Marking code OS doing by Shortblock mechanic</u> • Marking engine serial no. on cylinder block doing by Main Line mechanic																		
	Liner deck portion	See that Three Bond No.1215 (LG-6) has been applied to the proper area.																	
2	Cylinder liner	Crevice seal O-rings	Check that the chamfered side is facing down.																
			Check that the colour of the bottom O-ring is orange. Apply oil SAE 15W-40 immediately before installing.																
		Liner	- Set the "T" mark on the top surface of the liner facing the front. - Using a cloth, wipe off dirt and oil from contact surface of liner lip and cylinder block counter bore.																
		Liner Protrusion	Check the protrusion of the liner: <table border="1"> <tr> <th>Cyl. No</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> </tr> <tr> <td>Actual</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table>	Cyl. No	1	2	3	4	5	6	Actual							0.07 ~ 0.15 mm (Shop Manual page. 13-25)	
Cyl. No	1	2	3	4	5	6													
Actual																			
3	Crankshaft	Main Bearings	- Make sure that stamp mark on Crankshaft is available. - Check that there are no scratches or dents on the pin or main journal portion																
			Check size of Crank Shaft main bearings STD 0.25 0.50																
			Check that there is no dirt at the rear face of the bearing before tightening. Coat the inside face with engine oil.																
			- Do not coat the rear surface of the bearing - Upper bearing has oil hole - Lower bearing has no oil hole - Protrusion of Dowel Pin for Thrust. Actual: mm Make sure the number of main cap same with S/N Engine Block.	3.0 ~ 3.5 mm (Reman Standard)															
		Thrust bearing	- Install the thrust bearing with the groove on the crankshaft side or Main cap No.6. - Bolts must be oiled. When tightening, start from the center and work in order to the outside																
		Main Bearing Cap	1 st step torque&retorque Actual: kgm	29.0 ± 1.5 Kgm (Shop Manual page. 13-27)															
2 nd step torque&retorque Actual: kgm	58.0 ± 1.0 Kgm (Shop Manual page. 13-27)																		
3 rd step torque Actual: °	90° + 30° - 0 (Shop Manual page. 13-27)																		
- After tightening bolt, make punch mark on the head of each bolt. (Don't make a punch mark when using a new bolt) - Punch mark max. 5 punch marks.																			

ASSEMBLY CHECKSHEET ENGINE ASSY KOMATSU ENGINE 6D170E-5	WO NUMBER		DOCUMENT NO	RC/QR04/EG/02/03
	ENGINE MODEL		PUBLIC DATE	
	ENGINE SERIAL NO.		REVISION NO.	08
	MACHINE MODEL		PAGE	

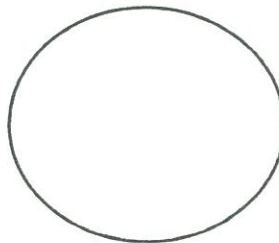
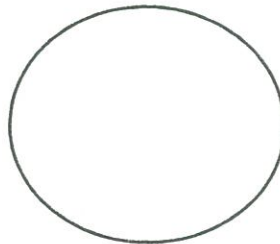
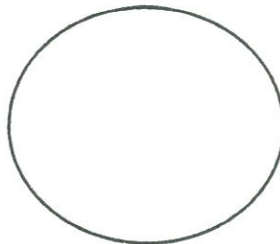
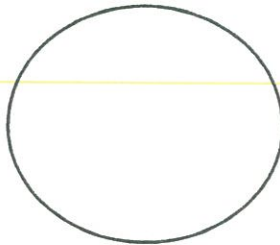
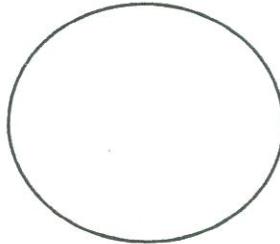
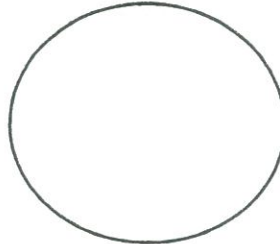
NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	ACTION DATE	SIGN		SKETCH DRAWING AND PHOTOES																					
						MEC	GL/QA																						
	CRANKSHAFT	Seal Surface	Clean of all dirt and dust on the contact surface of the front and rear seals																										
		End Play	Measure crankshaft end play with dial gauge Tolerance : Actual: mm	0.14 ~ 0.32 mm (Shop Manual page. 13-27)																									
			After tightening bolts, check that shaft turns smoothly. Turning torque actual = kgm	2.7 kgm or Less (Reman Standard)																									
		Face and Radial out of Crankshaft	Measure Face & Radial out of Crankshaft at Front and Rear surface Crankshaft before install Conrod assembly																										
			Rear Radial out: mm																										
			Rear Face Run out mm																										
			Front Radial out: mm																										
			Front Face Run out mm																										
				Maximum 0.05 mm(STD alignment Main bearing bore)																									
4	Piston and	Piston	- Set up each ring with the stamped mark facing up. Note the positioning of the ring gap. - For the oil ring, take out the expander and install it onto the piston, then install the oil ring - Confirm that the expander is properly fitted to the groove.																										
		Snap ring	Confirm that the snap ring which secures the piston pin is properly fitted to the groove. Check that the connecting rod moves smoothly																										
		Con. rod Bearing	- Check size of main bearings STD 0.25 0.50 - Match the number of conrods and conrods cap Check that there is no dirt at the rear face of the bearing before assembling. <table><tr><td>CYL</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Up</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Low</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> - Do not coat the rear surface of the bearing with oil.	CYL	1	2	3		4	5	6	Up							Low							STD or US 0.25 US. 0.50			
CYL	1	2	3	4	5	6																							
Up																													
Low																													
		Inserting Piston and Connecting Rod	- Check that the hole in the upper bearing and the oil hole in the con. rod are aligned. - Set the cylinder block on its side so that the assembly does not enter the cylinder suddenly when inserted. - Coat the cylinder liner with engine oil (SAE 15W-40) uniformly around the whole circumference by hand - Check that number of piston and connecting rod assembly match cylinder number. - Coat connecting rod lower bearing with engine oil, spreading over whole part with finger. - Conrod bolts are to have oil (SAE 15W-40) on the threads and under the washers.																										
		Big End Torque	1st step: (tighten in order shown on diagram) and Retorque Actual: kgm 2nd step: Repeat above step Actual: kgm - After tightening bolt, make punch mark on the head of each bolt. (Don't make a punch mark when using a new bolt) - Punch mark max. 5 punch marks. - Measure connecting rod side clearance . <table><tr><td>Cyl. No</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Actual</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Cyl. No	1	2	3	4	5	6	Actual							20.0 ± 1.0 kgm (Shop Manual Page. 13-29) 90°+30°-0 (Shop Manual Page. 13-29)											
Cyl. No	1	2	3	4	5	6																							
Actual																													
		Piston Position	- Make sure that piston position is in correct position. (see from top position) - Top Piston with deep notch is sitting to intake side <table><tr><td>Cyl. No</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Actual</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Cyl. No	1	2	3	4	5	6	Actual							0.20 ~ 0.374mm (Shop Manual Page. 13-30)											
Cyl. No	1	2	3	4	5	6																							
Actual																													
								      																					

ASSEMBLY CHECKSHEET ENGINE ASSY KOMATSU ENGINE 6D170E-5	WO NUMBER		DOCUMENT NO	RC/QR04/EG/02/03
	ENGINE MODEL		PUBLIC DATE	
	ENGINE SERIAL NO.		REVISION NO.	08
	MACHINE MODEL		PAGE	

NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	ACTION DATE	SIGN		SKETCH DRAWING AND PHOTOS														
						MEC	GL/QA															
5	Piston Cooling Nozzle	Piston Cooling Nozzle	Check that the piston cooling nozzle is not clogged and that the angle is correct. Use thread tightener (LT-2) on the mounting bolt. Actual: kgm	2.8 ~ 3.5 Kgm (Shop Manual Page. 13-30)				 														
			- Each Piston have two (2) piston cooling nozzles. Both of piston cooling nozzles is different physic and different code Cooling nozzle single hole (code F) must be installed a front side of cylinder Cooling nozzle double holes (code R) must be installed at rear side side of cylinder																			
			Please marking on the table																			
			<table><tr><td>Part Name</td><td>Liner</td><td>Piston</td><td>Pin Piston</td><td>Ring Piston</td><td>Conrod</td><td>Conrod Bolt</td></tr><tr><td>New</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Used</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>						Part Name	Liner	Piston	Pin Piston	Ring Piston	Conrod	Conrod Bolt	New						
Part Name	Liner	Piston	Pin Piston	Ring Piston	Conrod	Conrod Bolt																
New																						
Used																						
6	Cylinder Head	Mounting surface and Grommets	- Check that there is no dust or dirt inside the cylinder and on mounting surface of cylinder head. - When installing the gasket, check and mark grommet. <table><tr><td>Cyl.</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Act</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> - Grommet has been fitted and has't come out of position. - Don't coat grommet with gasket sealant.	Cyl.	1	2	3	4	5	6	Act							STD or OVER SIZE				 
			Cyl.	1	2	3	4	5	6													
		Act																				
		O/S head gasket	If oversize head gaskets are to be fitted, the block must be stamped O/S. <table><tr><td>STD</td><td>OS</td></tr><tr><td></td><td></td></tr></table>	STD	OS																	
STD	OS																					
Bolts	- The head bolts must have no dirt in the threads or under the washer, or in the threaded holes in the block. A tap can be used to clean the holes. - Coat the thread and washer of the bolts with anti seized thread (LM-P)/(Loctite 767)	<table><tr><td>7 Kgm</td></tr><tr><td>15 Kgm</td></tr><tr><td>25.0 ± 1 kgm (Shop Manual Page. 18)</td></tr><tr><td>39.0 ± 1 kgm (Shop Manual Page. 18)</td></tr><tr><td>90° ~120° (Shop Manual Page. 18)</td></tr><tr><td>9.5 ~ 10.5 kgm (Shop Manual Page. 18)</td></tr></table>	7 Kgm	15 Kgm	25.0 ± 1 kgm (Shop Manual Page. 18)	39.0 ± 1 kgm (Shop Manual Page. 18)	90° ~120° (Shop Manual Page. 18)	9.5 ~ 10.5 kgm (Shop Manual Page. 18)														
	7 Kgm																					
15 Kgm																						
25.0 ± 1 kgm (Shop Manual Page. 18)																						
39.0 ± 1 kgm (Shop Manual Page. 18)																						
90° ~120° (Shop Manual Page. 18)																						
9.5 ~ 10.5 kgm (Shop Manual Page. 18)																						
1 Mounting bolts 1 - 7 1st step and recheck: Actual: kgm 2nd step and recheck: Actual: kgm 3rd step and recheck: Actual: kgm 4th step and recheck: Actual: kgm 5th step and recheck: Actual: ° Target of 5th step=110° Mounting bolts 8 ~ 9 Actual: kgm																						
7	Cylinder Head Cont.		After tightening the bolts, make a punch mark an each on the head bolt (don't make a punch mark when using a new bolt					 														
			Punch mark maximum 3 punch marks (replace bolt if punch mark over than 3 punch marks)																			
			Make sure Installation heat label at cylinder head No.6																			
			99°C ~ 127°C (210°F~260°F)																			

MEASUREMENT LINER PROTRUSION EACH CYLINDER

FRONT ENGINE



CRITICAL ITEM CHECK

Name :	Name :
QA INSPECTOR	PROD. SUPERV SCR



SERVICES

PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

PISTON COOLING NOZZLE

INSPECTION CHECKSHEET

KOMATSU ENGINE 6D170-5 SERIES

MACHINE MODEL

WO NO:

SHORTBLOCK SECTION

CHECK SHEET GO NO GO PISTON COOLING NOZZLE

UNIT MODEL		DOCUMENT NO.	RC/QA/EG/252/R01
ENGINE MODEL		PUBLIC DATE	
COMP. S/N		REVISION NO.	00
WO.NO		PAGE	1 of 1

PROCESS	START		MECHANIC	
	FINISH		QA INSPECTOR	

Check Piston cooling Nozzle before used again
Using special tool GO NO GO (JIG), see picture as bellow

G	V
NG	X

Piston cooling nozzle marking on the top a head

SAA12V140E-3 V
SAA6D170E-3 F1,.....F6 ,R1,.....R6
SAA6D170E-5 F1,.....F6 ,R1,.....R6



SAA12V140E-3

LH

NO	RESULT	
	G	NG
V1		
V2		
V3		
V4		
V5		
V6		

RH

NO	RESULT	
	G	NG
V1		
V2		
V3		
V4		
V5		
V6		

SAA6D170E-3

NO	RESULT	
	G	NG
F1		
F2		
F3		
F4		
F5		
F6		

NO	RESULT	
	G	NG
R1		
R2		
R3		
R4		
R5		
R6		

SAA6D170E-5

NO	RESULT	
	G	NG
F1		
F2		
F3		
F4		
F5		
F6		

NO	RESULT	
	G	NG
R1		
R2		
R3		
R4		
R5		
R6		

CHECK BY

APPROVAL BY

MECHANIC

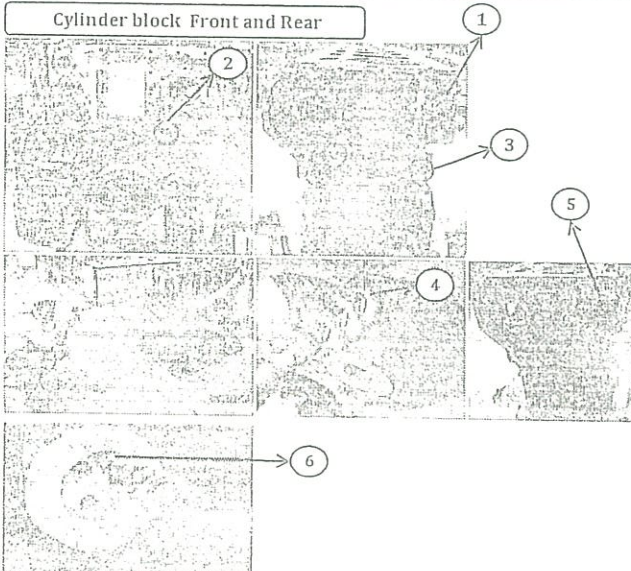
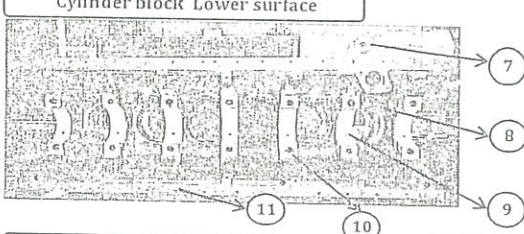
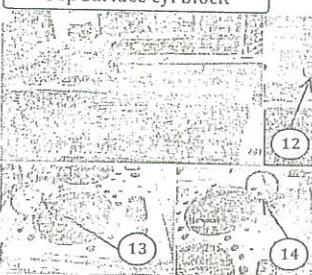
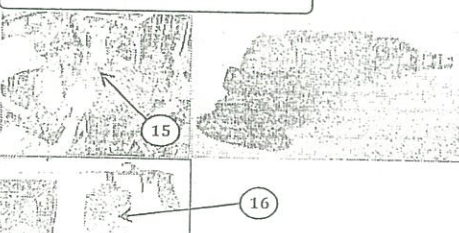
QA INSPECTOR



PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

CYLINDER BLOCK WASHING CHECKSHEET ENGINE

MACHINE MODEL		CYLINDER BLOCK P/N	6245-21-1100	DOCUMENT NO.	ENG-CB-6D170/3-2-00
ENGINE MODEL	6D170 SERIES	INSPECTION DATE		PUBL. DATE	29-DES-15
ENGINE SERIAL NO.		WO NUMBER		REVISION NO.	00
CUSTOMER NAME				PAGE	1 OF 1

SKETCH DRAWING OR PHOTOES	NO	ITEM CHECK	INSPECTION		REMARK
			OK	NOT OK	
Cylinder block Front and Rear 	1	Remove all plug on cyl block, specially on oil line gallery area. (Main gallery, piston cooling gallery, water gallery line)			
	2	Check all thread on seat plug area.			
	3	Clean Piston cooling gallery area use nylon brush Ø22mm and hand drill machine.			
	4	Clean main gallery area, sub gallery area and water line area use nylon brush Ø22mm and hand drill machine.			
	5	Clean oil line gallery on rear cyl block use nylon brush Ø22mm and hand drill machine.			
	6	Clean timing gear bolt on rear cyl block and oil line area use small nylon brush Ø8mm and hand drill machine.			
Cylinder block Lower surface 	7	Remove plug on big oil line gallery to oil pump, and clean use nylon brush Ø22mm and hand drill machine.			
	8	Clean oil line hole on piston cooling nozzle area and thread. Use nylon brush Ø8mm and hand drill machine.			
	9	Clean lubricating hole for all main journal (on cyl block), use nylon brush 8 mm and clean from lower until end of center main gallery.			
	10	Clean all thread for main cap bolt use nylon brush Ø22mm and make sure all of thread free from corrosion and dirty oil.			
Top surface cyl block 	11	Retap all mounting thread oil pan on cyl block use tap M10 and then clean use nylon brush Ø12mm.			
Clean surface contact area 	12	Clean all cyl head mounting thread (after retap) use nylon brush Ø22mm and dry use air.			
	13	Clean oil line lubricating to cyl head on top surface cyl block, use nylon brush Ø8mm.			
	14	Clean all contact surface on cyl block front, rear, RH, LH side area use wire brush and clean with washing chemical.			
	15	Coat new engine oil to all oil line gallery after clean and dry use air.			
	16				

* Special tap bolt is cyl head mounting bolt with modification.

Notes :

Checked by,

Inspected by,

Aproved by,

CRANKSHAFT WASHING CHECKSHEET ENGINE

MACHINE MODEL		CRANKSHAFT PIN		DOCUMENT NO.	ENG-CR-6D170/3-2-001
ENGINE MODEL	6D170 SERIES	WASHING DATE		PUBL. DATE	29-DES-15
ENGINE SERIAL NO.		WO NUMBER		REVISION NO.	00
CUSTOMER NAME				PAGE	1 OF 1

SKETCH DRAWING OR PHOTOS	NO	ITEM CHECK	INSPECTION		REMARK
			OK	NOT OK	
	1	Check condition crankshaft assy from corrosion and abnormality scratch on all main and pin journal before washing.			
	2	Recheck all thread condition, use FRONT DAMPER BOLT for check condition of thread.			
	3	Clean all thread use nylon brush (22 mm) and washing chemical until none dirty oil on all thread.			
	4	Recheck all front thread condition use FLYWHEEL BOLT for check all thread condition.			
	5	Clean gear from corrosion and dirty, use nylon brush or wire brush and washing chemical.			
	6	Clean all front thread use nylon brush (22 mm) use washing chemical.			
	7	Clean all main journal use nylon brush (8 mm) with extension. Until lubricating hole clean and none dirty oil.			
	8	Clean all pin journal use nylon brush (8 mm) with extension. Until lubricating hole clean and none dirty oil.			
	9	Clean Pin journal lubricating hole cross until main journal lubricating hole (distance L), for distance see drawing " B "			
* Lubricating hole on pin journal for engine 12V140 series have 2 holes, and all engine 6D140, 6D170, 6D125 series have 1 hole.					
Notes :					

Checked by,

Inspected by,

Approved by,

CONNECTING ROD WASHING CHECKSHEET

ENGINE

MACHINE MODEL		CONNECTING ROD P/N		DOCUMENT NO.	ENG-CR-6D170/3-2-001
ENGINE MODEL	6D170 SERIES	WASHING DATE		PUBL. DATE	29-Dec-15
ENGINE SERIAL NO.		WO NUMBER		REVISION NO.	00
CUSTOMER NAME				PAGE	1 OF 1

✓: GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

[illegible]

* For all connecting rod all engine series use nylon brush with modification handle spec Ø8mm for cleaning lubricating hole.

Notes :

Checked by,

Inspected by,

Aproved by,



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PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

CYL BLOCK

MEASUREMENT CHECKSHEET

KOMATSU ENGINE 6D170-5 SERIES

MACHINE MODEL

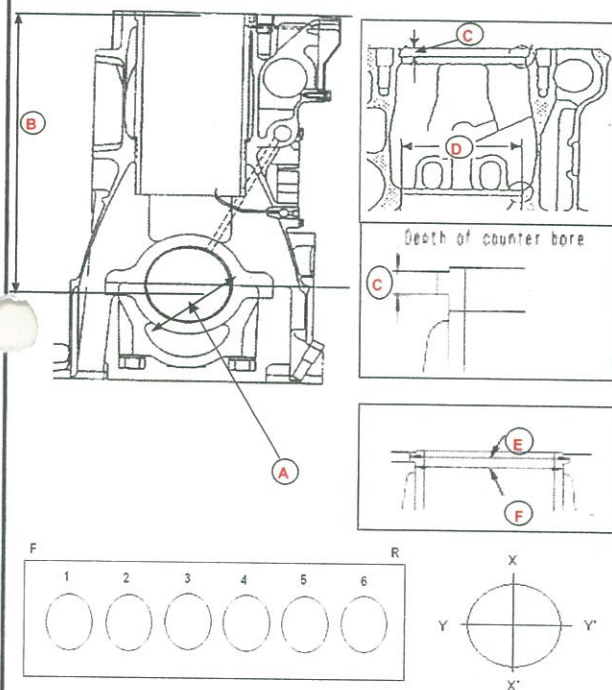
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ENGINE ASS'Y SECTION

CYLINDER BLOCK MEASURING INSPECTION CHECKSHEET

JOB NUMBER		PART NAME	CYLINDER BLOCK	DOCUMENT NO.	RC/QR03/EG/01/03
ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	

SKETCH DRAWING



MACHINIST NAME

Cyl. Block Inspection

Machining Process

Standard Size

Unit : mm

DESCRIPTION	STANDARD SIZE
A MAIN BEARING BORE	147.999 ~ 148.025
B CYLINDER BLOCK HEIGHT	488.960 ~ 489.040
USE OVER SIZE GASKET	488.560 ~ 488.960
C COUNTER BORE DEPTH	14.000 ~ 14.050
D INSIDE DIAMETER(AT O-RING LAND)	190.290 ~ 190.560
E INSIDE DIAMETER(AT FLANGE SEAT)	205.150 ~ 206.025
F INSIDE DIAMETER (AT COUNTERBORE PORTION)	194.430 ~ 194.590

TIGHTENING TORQUE OF MAIN BEARING CAP

1 st step	=	27.5 ~ 30.5 Kgm
2 nd step	=	57.0 ~ 59.0
3 rd step	=	90° ~ 120°

VISUAL CHECK

OK

Not OK

CRACK HAS BEEN CHECKED

MANHOURS UTILIZATION

	START	FINISH	MAN HOURS
Measuring Process			
TOTAL			

INSPECTION SALVAGE PROCESS BY MACHINIST

RESULT : ☒ GOOD CONDITION

X : BAD CONDITION

MAIN BEARING BORE

Main Bore No.		X - X'	Y - Y'	Z - Z'
1	A			
	B			
2	A			
	B			
3	A			
	B			
4	A			
	B			
5	A			
	B			
6	A			
	B			
7	A			
	B			

CYLINDER BLOCK MEASURING INSPECTION CHECKSHEET

JOB NUMBER		PART NAME	CYLINDER BLOCK	DOCUMENT NO.	RC/QR03/EG/01/03
ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	

C	COUNTER BORE DEPTH	14.000 ~ 14.050
D	INSIDE DIAMETER (AT O-RING LAND)	190.290 ~ 190.560

COUNTER BORE DEPTH

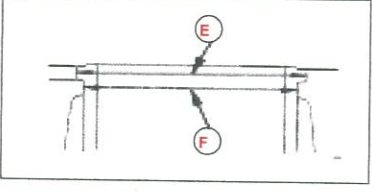
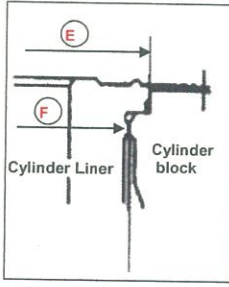
BEFORE MACHINING												
Cyl	X			X'			Y			Y'		
	MECH	QA	GL	MECH	QA	GL	MECH	QA	GL	MECH	QA	GL
1												
2												
3												
4												
5												
6												
AFTER MACHINING												
Cyl	X			X'			Y			Y'		
	MECH	QA	GL	MECH	QA	GL	MECH	QA	GL	MECH	QA	GL
1												
2												
3												
4												
5												
6												

INSIDE DIAMETER (AT O-RING LAND)

D	Cyl.	No.	MECH		QA/GL	Cyl.	No.	MECH		QA/GL
			X	Y				X	Y	
		1					4			
		2					5			
		3					6			

CYLINDER BLOCK MEASURING INSPECTION CHECKSHEET

JOB NUMBER		PART NAME	CYLINDER BLOCK	DOCUMENT NO.	RC/QR03/EG/01/03
ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	

CYLINDER BLOCK (AT FLANGE SEAT)			
 	E	INSIDE DIAMETER (AT FLANGE SEAT)	205.150 ~ 206.025
	F	INSIDE DIAMETER (AT COUNTERBORE PORTION)	194.430 ~ 194.590
Note :			

INSIDE DIAMETER (AT FLANGE SEAT)

E	X - X'				Y - Y'			
	Cyl	CHECK			Cyl	CHECK		
		MEC	QA/GL	SPV		MEC	QA/GL	SPV
	1				1			
	2				2			
	3				3			
	4				4			
	5				5			
	6				6			

INSIDE DIAMETER (AT COUNTERBORE PORTION)

F	X - X'				Y - Y'			
	Cyl	CHECK			Cyl	CHECK		
		MEC	QA	SPV		MEC	QA	SPV
	1				1			
	2				2			
	3				3			
	4				4			
	5				5			
	6				6			

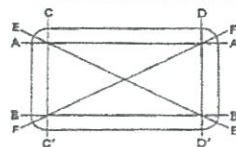
CYLINDER BLOCK MEASURING INSPECTION CHECKSHEET

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ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	

Judgement of Distortion (Flatness) : Measure thickness between Block & Straight edge.

Flatness	Remedy
within 0.08mm	Correct with surface grinding

Measurement tool : Straight edge, clearance gauge



THICKNESS BETWEEN BLOCK AND STRAIGHT EDGE

CYLINDER	A - A'	B - B'	C - C'	D - D'	E - E'	F - F'
1						
2						
3						
4						
5						
6						

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVAL BY

MECHANIC(MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION



SERVICES

PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

ENGINE ASS'Y

MAINLINE ASSEMBLY

CHECK SHEET

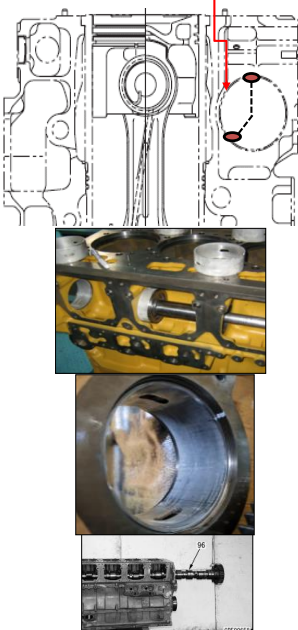
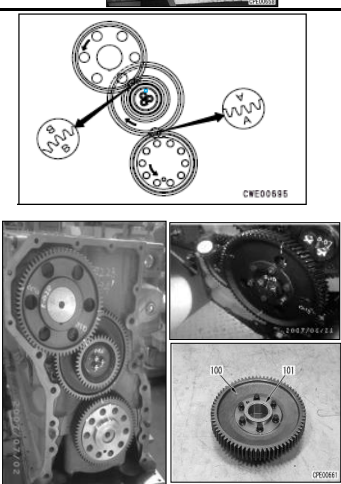
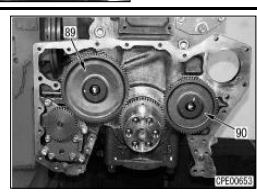
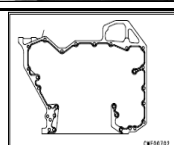
KOMATSU ENGINE 6D170-5

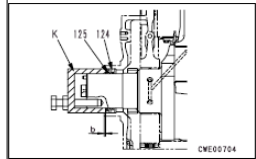
MACHINE MODEL

WO NO:

SHORTBLOCK SECTION

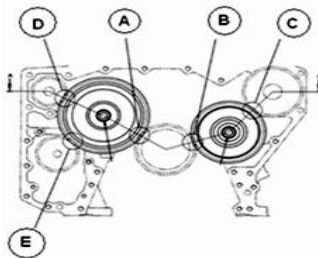
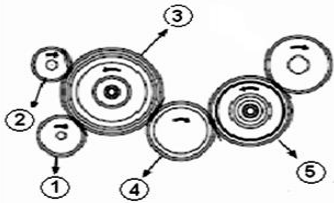
ASSEMBLY CHECKSHEET ENGINE ASSY KOMATSU ENGINE 6D170E-5	WO NUMBER		DOCUMENT NO	RC/QR04/EG/02/03
	ENGINE MODEL		PUBLIC DATE	
	ENGINE SERIAL NO.		REVISION NO.	08
	MACHINE MODEL		PAGE	

NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	ACTION DATE	SIGN		SKETCH DRAWING AND PHOTOS																																
						MEC	GL/QA																																	
6	Camshaft	Assembly	- Check camshaft bushes are correctly installed in block, and oil supply holes are aligned. - Check aligned oil supply holes in bushing camshaft Oil hole mustbe visible at least min 60% <table><tr><th>No</th><th>Bushing cam No</th><th>Oil supply hole</th><th>Connector bushing facing up</th></tr><tr><td>1</td><td>1</td><td></td><td></td></tr><tr><td>2</td><td>2</td><td></td><td></td></tr><tr><td>3</td><td>3</td><td></td><td></td></tr><tr><td>4</td><td>4</td><td></td><td></td></tr><tr><td>5</td><td>5</td><td></td><td></td></tr><tr><td>6</td><td>6</td><td></td><td></td></tr><tr><td>7</td><td>7</td><td></td><td></td></tr></table>	No	Bushing cam No	Oil supply hole	Connector bushing facing up	1	1			2	2			3	3			4	4			5	5			6	6			7	7							sambungan bushing menghadap atas engine 
No	Bushing cam No	Oil supply hole	Connector bushing facing up																																					
1	1																																							
2	2																																							
3	3																																							
4	4																																							
5	5																																							
6	6																																							
7	7																																							
			- When installing the camshaft assembly, rotate it to protect the cam bushing from damage. - Make sure Camshaft can rotated smoothly - Match the marks B on the camshaft gear and small idler gear - Mounting bolt of camshaft plate: Actual: kgm - Measure the camshaft end play Actual: mm	10.0 ~ 12.5 kgm (Shop Manual Page. 13-32) 0.05 ~ 0.20 mm (Shop Manual Page. 13-32)																																				
8	Rear Idler gear		- Make sure all gears are clean and no excessive pitting - Install large gear and small gear - Tightening torque bolt M12 x 1.75 Actual: kgm - Spacer with V-Groove machined surface facing to the Cylinder block. - Assure shaft and bushing of gears are normal when be rotated - Set the (UP) mark on the shaft facing up - Make sure mark A on the crankshaft gear and large idler gear are match. - Mounting bolt 1st step Actual: kgm - Mounting bolt 2nd step Actual: °	10.0 ~ 12.5 kgm (Shop Manual Page. 13-30) 5.8 ± 1.2 kgm (Shop Manual Page. 13-31) 90° ~120° (Shop Manual Page. 13-31)																																				
9	Front Idler Gear		- Make sure gears, bushes are good condition - Make sure that shaft can be rotate in normal condition. - Idler gear (small and Large) mounting bolt: Actual: kgm	25.0 ~ 31.5 kgm (Shop Manual Page. 13-34)																																				
10	Oil Pump		- Check all oil pump condition - Fill in oil engine to inner of pump - Rotate pump by hand smoothly - Fit O-Ring to sensor side - Mounting oil pump and tighten mounting bolts Actual: kgm <table><tr><th colspan="2">OIL PUMP</th></tr><tr><th>NEW</th><th>REUSE</th></tr><tr><td></td><td></td></tr></table> - Recheck plug of main relief valve (retighten) - Check end play Oil pump gear Actual: mm <table><tr><th colspan="2">RELIEF VALVE</th></tr><tr><th>NEW</th><th>REUSE</th></tr><tr><td></td><td></td></tr></table>	OIL PUMP		NEW	REUSE			RELIEF VALVE		NEW	REUSE			7Kg (Reman Standart) 0.04 ~ 0.18 mm (Reman Standard)																								
OIL PUMP																																								
NEW	REUSE																																							
RELIEF VALVE																																								
NEW	REUSE																																							
11	Front Cover		- No crack on front cover - No damage threads - Coat the gasket sealant as shown in diagram with a line of width 2 ~ 3 mm (LG-6/Three bond1215)																																					

ASSEMBLY CHECKSHEET ENGINE ASSY KOMATSU ENGINE 6D170E-5		WO NUMBER				DOCUMENT NO		RC/QR04/EG/02/03								
		ENGINE MODEL				PUBLIC DATE										
		ENGINE SERIAL NO.				REVISION NO.		08								
		MACHINE MODEL				PAGE										
12	Front Seal		- Make sure no scratches on crankshaft (seal contact area) - Check visual of front seal and make sure that seal oil good condition and no scratch on seal lip area.			STD or OVER SIZE S = Single Lip W = Double Lips										
			<table border="1" style="display: inline-table; border-collapse: collapse;"> <tr> <td style="width: 33%;">S/W</td> <td style="width: 33%;">STD</td> <td style="width: 33%;">OS</td> </tr> <tr> <td style="height: 20px;"></td> <td></td> <td></td> </tr> </table>		S/W	STD	OS									
S/W	STD	OS														

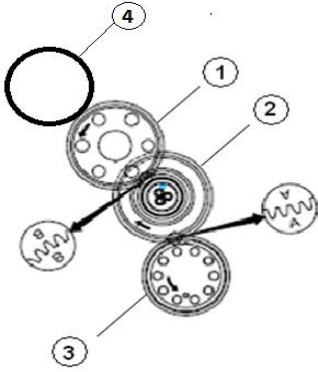
FRONT SIDE TIMING GEAR

Timing Gears Marked	Check Point	Backlash (mm)	Measured	ACTION DATE	SIGN	
					MEC	GL/QA
A	Crankshaft gear and Idle gear (large)	0.134 ~ 0.326 (R/L: 0.5)				
B	Crankshaft gear and Idle gear (Medium)	0.114 ~ 0.320 (R/L: -)				
C	Idle gear (Medium) and Fuel pump drive gear	0.114 ~ 0.320 (R/L: 0.5)				
D	Idle gear (Large) and Water pump drive gear	0.121 ~ 0.333 (R/L: 0.5)				
E	Idle gear (Large) and Oil pump drive gear	0.121 ~ 0.333 (R/L: 0.5)				
No.	Check Point	End play (mm)	Measured	ACTION DATE	SIGN	
					MEC	GL/QA
1	Oil pump drive gear	0.04 ~ 0.18				
2	Water pump drive gear	0.04 ~ 0.20				
3	Idle gear	0.04 ~ 0.17				
4	Crankshaft gear	0.14 ~ 0.32				
5	Idle gear	0.04 ~ 0.17				

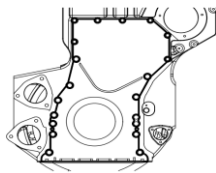
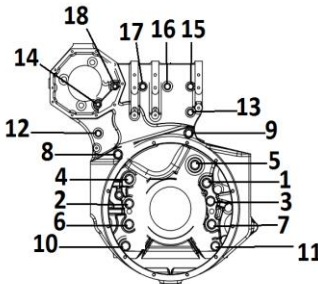
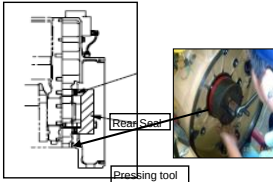
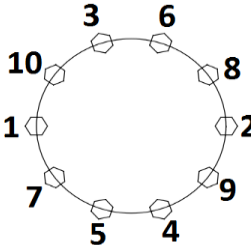
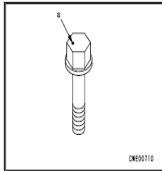
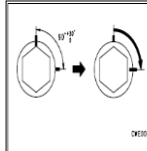
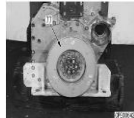
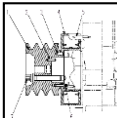



REAR SIDE TIMING GEAR

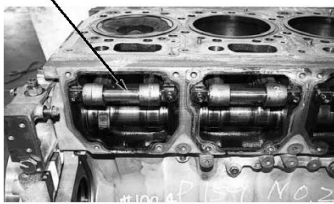
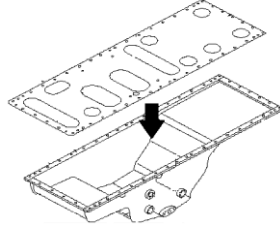
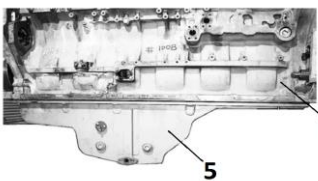
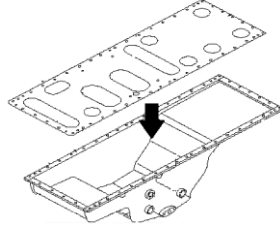
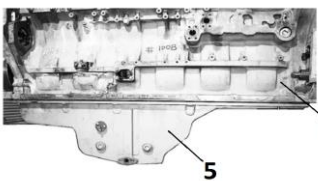
Timing Gears Marked	Check Point	Backlash (mm)	Measured	ACTION DATE	SIGN	
					MEC	GL/QA
A	Crankshaft gear and Idle gear (large)	0.155 ~ 0.412 (R/L: 0.6)				
B	Camshaft gear and Idle gear (Medium)	0.145 ~ 0.380 (R/L: 0.6)				
No.	Check Point	End play (mm)	Measured	ACTION DATE	SIGN	
					MEC	GL/QA
1	Camshaft drive gear	0.05 ~ 0.20				
2	Idle gear	0.04 ~ 0.17				
3	Crankshaft drive gear	0.14 ~ 0.32				
4	Supply pump drive gear	0.25 ~ 0.30				



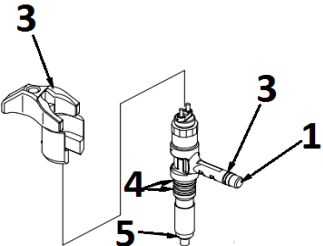
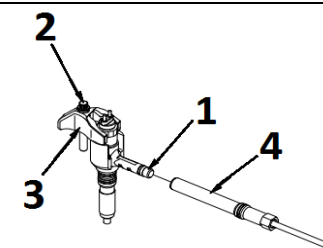
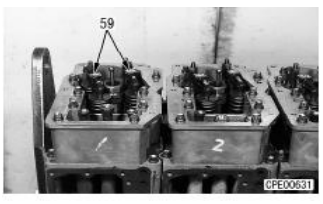
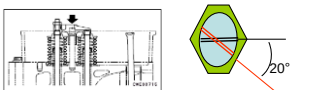
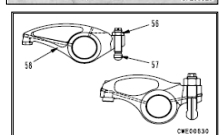
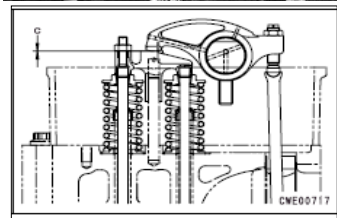
ASSEMBLY CHECKSHEET ENGINE ASSY KOMATSU ENGINE 6D170E-5	WO NUMBER		DOCUMENT NO	RC/QR04/EG/02/03
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NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	ACTION DATE	SIGN		SKETCH DRAWING AND PHOTOES							
						MEC	GL/QA								
13	Flywheel Housing		- No crack and no damage - No damage thread hole Coat with gasket sealant (LG-7/Three bond1207D), then raise flywheel housing and install.					 							
		- Tighten the mounting bolt with specified order: 1st step Actual: kgm	36 ~ 40 kgm (Shop Manual Page. 13-36)												
		2nd step Actual:	Loosen (Shop Manual Page. 13-36)												
		3rd step Actual: kgm	40 ~ 44 kgm (Shop Manual Page. 13-36)												
		(tighten the main mounting bolts in the specified order)													
		Other bolts M12 x 1.75 Actual: kgm	10.0 ~ 12.5 kgm (Shop Manual Page. 13-36)												
	Rear Seal		- Make sure that the rear seal , no pitting, no scratch - Press fit the rear seal by use installation tool to obtain correct position. - Be careful not to damage the seal when installing the housing.	<table><tr><td>SW</td><td>STD</td><td>OS</td></tr><tr><td></td><td></td><td></td></tr></table> STD or OVER SIZE S = Single Lip W = Double Lips	SW	STD	OS								
SW	STD	OS													
14	Flywheel		- Make sure flywheel is not crack or damage of thread - Coat thread and flange bolts with engine oil					   							
		- Tighten the Flywheel mounting bolts in the specified order. 1st step: Actual: kgm	20 ±1 kgm (Shop Manual Page. 13-37)												
		2nd step: Actual: kgm	47 ± 2 kgm (Shop Manual Page. 13-37)												
		3rd step: Actual: °	Tighten 90°⁺¹⁵₋₀ (Shop Manual Page. 13-37)												
		After tighten the bolts, make 1 punch mark on the head of each bolt (don't make a punch mark when using a new bolt). Mounting bolts punch are mark are not more than 5 punch marks													
		- Install engine speed sensor - Screw in until the tip of the sensor contacts the ring gear, then turn the sensor back	1/2 ~ 3/4 turn (Shop Manual Page. 13-38)												
		Tightening lock nut of the sensor Actual: kgm	3.5 ~ 4.8 kgm (Shop Manual Page. 13-38)												
15	Vibration Damper		Check that the vibration damper has been tested or renewed before installing.(no crack and no damage)					  <table><tr><td>1. Crankshaft pulley</td></tr><tr><td>2. Pin (crankshaft – crankshaft pulley)</td></tr><tr><td>3. Pin (crankshaft – vibration damper)</td></tr><tr><td>4. Vibration damper</td></tr><tr><td>5. Pointer</td></tr><tr><td>6. Bolt (crankshaft – vibration damper)</td></tr><tr><td>7. Bolt (crankshaft – crankshaft pulley)</td></tr></table>	1. Crankshaft pulley	2. Pin (crankshaft – crankshaft pulley)	3. Pin (crankshaft – vibration damper)	4. Vibration damper	5. Pointer	6. Bolt (crankshaft – vibration damper)	7. Bolt (crankshaft – crankshaft pulley)
		1. Crankshaft pulley													
		2. Pin (crankshaft – crankshaft pulley)													
		3. Pin (crankshaft – vibration damper)													
		4. Vibration damper													
		5. Pointer													
6. Bolt (crankshaft – vibration damper)															
7. Bolt (crankshaft – crankshaft pulley)															
Coat the thread and washer of the bolts with engine oil (SAE15W-40)															
Tighten damper mounting bolts to Pulley. Actual: kgm	6.7±0.7Kg (Reman Standard)														
- Tighten the crankshaft mounting bolts in the order shown. 1st step: Actual: kgm	7.5 ± 2 kgm (Shop Manual Page. 13-39)														
2nd step: Actual: kgm	25 ± 2 kgm (Shop Manual Page. 13-39)														
3rd step: Actual: kgm	65 ± 2 kgm (Shop Manual Page. 13-39)														

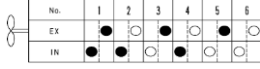
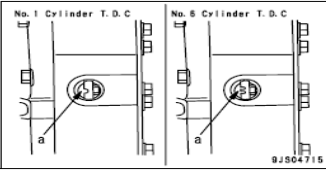
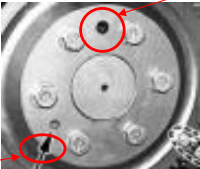
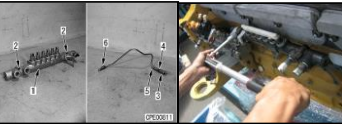
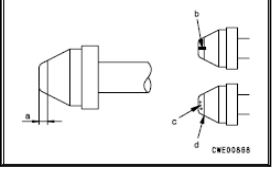
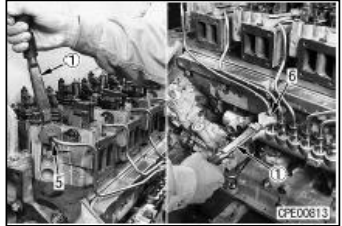
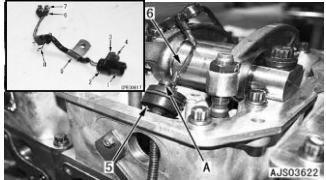
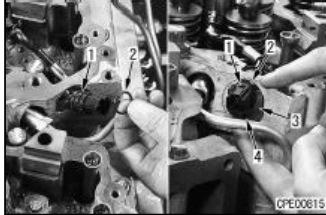
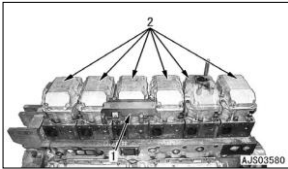
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NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	ACTION DATE	SIGN		SKETCH DRAWING AND PHOTOES
						MEC	GL/QA	
16	Cam Follower	assembly	Before installing the cam follower assembly, wash the oil hole of the mounting bolt, then immerse whole assembly in engine oil (SAE 15W-40).					  Cast Iron Allumunium
			Recheck cam follower(1) Rotate all of roller cam and make sure rotating of the roller cams are normal					
			Set the cam follower until can be installed Torque the cam follower mounting bolt : Actual: kgm	9.2 ~ 10.6 kgm (Shop Manual Page. 17)				
			Check that side of the cam follower shaft with the ball knocked in is facing the front of the engine.					
17	Oil pan	Cover	- Install the O ring than install cam follower cover. - Install the cast iron made cover no 2 and no 6. torque mounting bolt: Actual: kgm	6.7 ± 0.7 kgm (Shop Manual page. 00-12)				   
			Install the allumunium made cover no 1,3,4 and 5. torque mounting bolt Actual: kgm	1.1 ± 0.1 kgm (Shop Manual Page. 17)				
			Recheck condition of under plate from dent, and damage.					
			Coat with gasket sealant (LG-7/Threebond 1207D) and than install under plate - Coat the gasket sealant as shown in the diagram with a line of width Ø 2 ~ 3 mm - Temporarily tighten 4 bolts on the inside and several bolts on the outside untill finally installing the suction tube and oil pan					
18	Rocker Arm Housing	Suction tube and Brackets	- Check oil pump suction tube are clean, no crack, no rust, no dust. - Fit O-Ring and install suction tube and brackets Tighten the mounting bolts Actual: kgm	6.7±0.7Kgm (Shop Manual Page. 00-12)				   
			Install brackets Determine the position of the suction tube and brackets, then tighten the mounting bolts After installing the suction tube, remove all the temporary bolts (inside, outside) of the under plate					
			Check that the oil pan is not crack, no bend, no damage and clean					
			Coat the oil pan with liquid (LG-7/Threebond 1207D) - Coat the gasket sealant as shown in the diagram with a line of width Ø 2 ~ 3 mm - Install the oil pan and when tightening, start from center and work in order to the outside - Tighten mounting bolts of the oil pan Actual: kgm	6.7±0.7Kgm (Shop Manual Page. 00-12)				
19	Push rod		Check the tip of the push rod is fitted securely into the socket in the cam follower (push rods for the injectors and the valves)					 
			Make sure that the push rod position is correct					

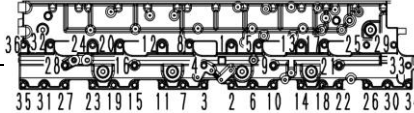
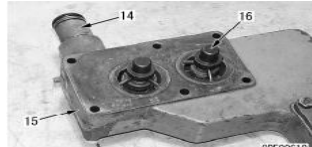
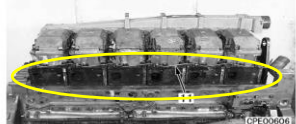
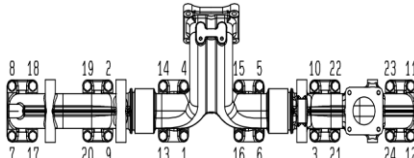
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						MEC	GL/QA																		
20	Injector	1	Check that the inside of the injector sleeve is free from dirty.																						
Install Gasket and oring and to grove to the center.  O-ring: Engine oil (EO30)																									
Insert injector to adapter and injector holder, and then insert them to cyl head while direction piping injector connector. Check all injector before assembly. <table><tr><td>Cyl</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Check</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>			Cyl	1	2	3	4		5	6	Check														
Cyl			1	2	3	4	5		6																
Check																									
- Write injector code below: <table><tr><th>CYL.NO</th><th>INJECTOR CODE</th><th>HOLE OF NOZZLE</th></tr><tr><td>1</td><td></td><td></td></tr><tr><td>2</td><td></td><td></td></tr><tr><td>3</td><td></td><td></td></tr><tr><td>4</td><td></td><td></td></tr><tr><td>5</td><td></td><td></td></tr><tr><td>6</td><td></td><td></td></tr></table> <i>*Example : Injector code 6290 05M 0128</i> <i>Hole of nozzle: 8 Holes</i> - Fit ball washer to bolt and tighten injector holder. - Torque mounting bolt holder: Engine oil (EO30) Actual: kgm	CYL.NO	INJECTOR CODE	HOLE OF NOZZLE	1			2			3			4			5			6			6.0 ~ 7.5 kgm (Shop Manual Page. 20)			
CYL.NO	INJECTOR CODE	HOLE OF NOZZLE																							
1																									
2																									
3																									
4																									
5																									
6																									
After tightening the injector holder ,remove the high pressure piping and install the other injector.																									
Measure the length a of the portion below the bolt head, If it is more than 80 mm, do not reuse the bolt; replace it with a new part. Actual : 0																									
21	Cross head		Make sure that the cross heads are good condition					 																	
Check that cross head can be moved up-down normally																									
Screw/nut position is in exhaust side.																									
- Adjust cross head securely when the screw has touched the valve stem, tighten adjustment screw a further 20° ~ 30° - Secure in position with locknut Actual: kgm			5.4 ~ 6.6 kgm (Shop Manual Page. 22)																						
22	Rocker Arm Ass'y	Rocker Arm	Make sure the screw and rocker arm thread good condition					 																	
Screw in adjustment screw until ball portion contact arm																									
Install intake, injector and exhaust arm to shaft Insert into the arm so that the inside of the shaft with the rivet knocked in faces the front of the engine and then coat with engine oil																									
Install rocker arm assemblies so that the shaft hole with the large inside diameter faces the rocker arm housing.																									
23	Valve mechanism and injector	Adjustment Valve Clearance and Injector	Tighten of the rocker arm mounting bolt Actual: kgm	24 ~ 26 kgm (Shop Manual Page. 23)				 																	
After installing rocker arm assembly make sure the arrangement is correct. Step and method adjustment of value clearance following to the shop manual is attached.																									
- Firing order 1 – 5 – 3 – 6 – 2 – 4 - Valve clearance : Intake valve Actual: mm			0.32 ± 0.02 mm (Shop Manual Page. 24)																						
Exhaust valve Actual: mm			0.62 ± 0.02 mm (Shop Manual Page. 24)																						

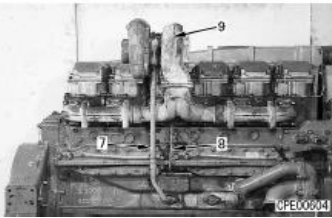
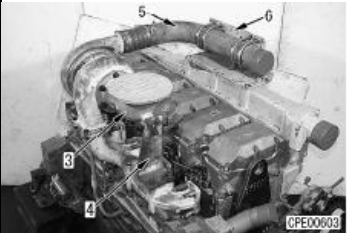
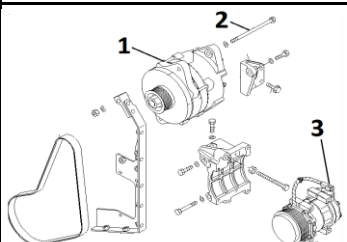
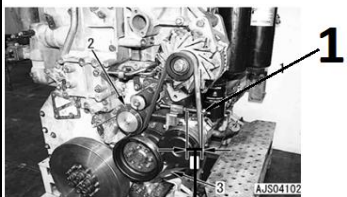
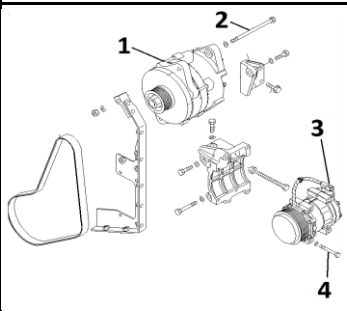
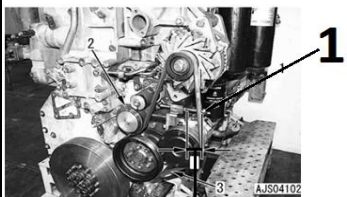
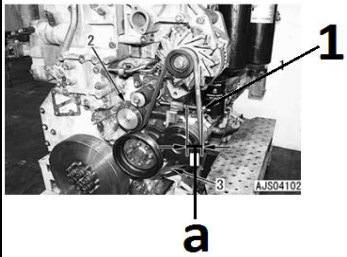
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						MEC	GL/QA	
24	Valve mechanism		- Adjustment injector table :  - Make sure all of bolts and nuts have tightened completely - Tightening torque lock nut of valve rocker arm. Actual: kgm	5.9 ~ 7.9 kgm (Shop Manual Page. 24)				
25	Fuel supply Pump	1	Install fuel supply pump by performing the removal procedure in reverse order. - Make sure that the stamed (C) of idle gear and suply pump drive gear are matched. - Make sure that knock pin inserted to knock pin hole on suply pump drive gear. - Tightening torque: Actual : Kgm - Tighten fuel supply pump drive gear. (bolt coat loctite) Actual : Kgm - Tightening fuel supply pump mounting bolts Actual : Kgm FUEL SUPPLY PUMP S/N: S/N: Note : Pastikan Lubang dowel tidak tertukar dengan lubang baut untuk treacker.	Loctite 262 6.7 ± 0.7 kgm (Shop Manual page. 13 32) 6.7 ± 0.7 kgm (Shop Manual page. 00 12)				 
	Fuel Pump, Common Rail Injector and Pipes.	Common rail and High press. Pipe	Mounting common rail to the Cylinder block and tightening the mounting bolt Torque : Actual : Kgm Assembly common rail High pressure pipe. - Fit O-Ring to high-pressure pipe sleeve on the injector side. - Tightening the sleeve nut with torque Actual : Kgm - Torque clamp mounting bolt Actual : Kgm - When installing the high pressure pipe, check it for visible longitudinal slip "b", pit "c" or level difference (wear) in part "d" (End of the taper seal)	4 ~ 5 kgm (Shop Manual page. 13 42) 5 ~ 6 kgm (Shop Manual page. 13 42) 6.0~ 7.5 kgm (Shop Manual page. 13 39) 4 ~ 5 kgm (Shop Manual page. 13 39) 1.0 ± 0.1 kgm (Shop Manual page. 13 43) Part "a" 2 mm from the end				  
		Injector	Apply engine oil to the spherical surface of the ball washer (to fit the ball washer) Tightening the Holder mounting bolt Actual : Kgm	6.7 ± 0.7 kgm (Shop Manual page. 00 12)				
		Wiring harness	Install O-Ring wiring harness to the connector hole and apply engine oil (SAE 15W ~ 40). And then install wiring harness terminal to injector terminal and tighten 2 nuts and lock tightening terminal nut Standart torque terminal nut Torque : Actual : Kgm	0.18~0.22kgm (Shop Manual page. 13 40)				
26	Cylinder Head Cover		- Fit O-Ring and insdtall the covers - Tightening torque of covers mounting bolts No.1,3,4,5 & 6 Actual: kgm No 2 Actual: kgm	1 ± 0.1 kgm (Shop Manual Page. 25) 6 ~ 7.5 kgm (Shop Manual Page. 25)				

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27	Oil Level gauge	Oil filler	- Install oil pressure sensor. Install outer component as a position and tighten all of component specified. - Tighten gauge tube sleeve nut Actual: kgm Make sure the dipstick is according to the machine it is mounted on.	4.5 ~ 6 kgm (Shop Manual Page. 13-47)				
28	Intake Manifold Assembly	Intake manifold 1	- Fit o-ring and gasket correctly. - Install intake manifold When the tightening the mounting bolts, start in the center and move outwards on each side in turn Actual: kgm	6.75 ± 0.5 kgm (Reman Standard)				
		Outer parts	- Boost sensor Actual: kgm - Fit o-ring and install fuel rail tube Tighten of Eyebolt Actual: kgm - Tighten of Sleeve nut Actual: kgm - Fit o-ring and install timing rail tube Tighten of Eyebolt Actual: kgm - Tighten of Sleeve nut Actual: kgm - Fit o-ring and install fuel inlet tube Tighten of Sleeve nut Actual: kgm - Fuel Tube Tighten of Sleeve nut Actual: kgm	1.37 ± 0.08 kgm (Shop Manual Page. 13-48) 2.5 ~ 3.5 kgm (Shop Manual Page. 13-48) 2.4 ~ 2.8 kgm (Shop Manual Page. 13-48) 3.5 ~ 4.5 kgm (Shop Manual Page. 13-48) 4.4 ~ 4.8 kgm (Shop Manual Page. 13-48) 6.1 ~ 6.9 kgm (Shop Manual Page. 13-48) 9.2 ~ 9.7 kgm (Reman Standard)				 
29	Oil Cooler Assembly		- Check that there is no dust or dirt inside the oil cooler and no crack - Coat the oil cooler gasket with gasket sealant (three bond) and install the oil cooler ass'y to the cyl. block. Tighten mounting bolts : Act : Kgm	4.0 ~ 7.0 kgm (Reman Standard)				
30	Water Pump Assembly, Thermosta and Thermosta housing	Water pump	- Recheck condition of water pump WATER PUMP NEW ☐ REUSE ☐ - Fit O-Ring and correctly and install water pump assembly carefully The water connector can also be installed after installing the thermostat housing					
		Thermostat housing and Thermostat	- Fit O-Ring and install by pass tube to thermostat housing Push in the water connector until it contacts the inside of the thermostat housing. - Install seal thermostat correctly - Install 2 (two) thermostat carefully. Coat the gasket on both sides with gasket sealant with a line of width Ø 2 ~ 3 mm (LG-6) - Install thermostat housing completely and push down by pass tube then secure in position with stopper. Actual: kgm - Install water temp sensor and tightening Actual: kgm	6.7 ± 0.7 kgm (Reman Standard) 1.37 ± 0.08 kgm (Shop Manual Page. 13-51)				 
31	Air bleed tube & Exhaust Manifold	Install air bleed tube	Install air bleed tube and tighten of eye bolt Actual: kgm	1 ~ 1.3 kgm (Shop Manual Page. 13-51)				
		exh. manifold assembly	- Install the gasket so that the (out) stamp faces the outside of the engine. Coat thread of exh. mounting bolt with seizure prevention . LM-P (anti seize) - Tighten the mounting bolts in the specified order Order to tightening of exh. manifold 1st step Actual: kgm 2nd step Actual: kgm	4 ~ 6 kgm (Shop Manual Page. 13-52) 11.5 ~ 12.5 kgm (Shop Manual Page. 13-52)				

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32	Turbo-charger Assembly	Turbocharger	Recheck turbocharger condition and rotation. TURBO CHARGER NEW REUSE					 
			Fit gasket and install the turbocharger Coat thread mounting bolts with seizure prevention (LM-P or anti seize).	Except D375A-5 8 ~ 10 kgm D375A-5 only 6 ~ 7.5 kgm (Shop Manual Page. 13-52)				
			Tightening torque of the mounting bolt Actual: kgm					
			- Install lubrication inlet tube and lubrication outlet tube. - Tightening torque of Eye bolt for inlet tube. Actual: kgm Fill oil on oil inlet of Turbocharger for pre Lubricating when first running in Dyno(fill until fully on oil inlet hole)	2.5 ~ 3.5 kgm (Shop Manual Page. 13-52)				
33	Starting Motor	Connector	Install bracket and intake connector and tightening clamp bolt Actual: kgm	1.0 ± 0.1 kgm				
		1	Install bracket and exhaust connector Coat thread mounting bolt with seizure prevention and rust prevention lubricant (LM-P)					
			TURBOCHARGER ASSY Serial Number:					
			S/N:					
34	Alternator	Starting Motor	Fit gasket and install starting motor assembly and tightening torque the mounting bolt. STARTING MOTOR NEW REUSE Tightening torque : Actual: kgm The number of the starting motors differs according to the machine they are mounted on	20 kgm (Reman Standard)				 
		Wiring	Recheck the starting motor assembly and make sure no cable shorted and no broken something else.					
		1	- Recheck condition all of wiring. - No broken - No disinsulation - No broken at the socket. - Install all wiring. If something happen on wiring and socket, please repair first before continue. The large connectors for ECVA & ECM are secured with bolts					
			STARTING MOTOR ASSY (code)S/N:					
34	Alternator	Alternator	- Remove oil filter on the front side. Alternator NEW REUSE - Insert the mounting bolt in the rear side toward the front side. - Install Alternator. - torque mounting bolt alternator: Tightening torque : Actual: kgm	11.5 ~ 12.5 kgm				 
		1	ALTERNATOR ASSY (code)S/N:					
			S/N					
		Compressor	- Install compressor. - torque mounring bolt compressor: Tightening torque : Actual: kgm	2.8 ~ 3.5 kgm				
34	Alternator	belt auxiliary	- Install auxiliary equipment belt. - Loosen adjustment bolt and give tension of tensioner belt until dimension (a) is obtained. - Tension the lock nut of adjustment bolt(1).					

ASSEMBLY CHECKSHEET ENGINE ASSY KOMATSU ENGINE 6D170E-5	WO NUMBER		DOCUMENT NO	RC/QR04/EG/02/03
	ENGINE MODEL		PUBLIC.DATE	
	ENGINE SERIAL NO.		REVISION NO.	08
	MACHINE MODEL		PAGE	

Carry out a visual inspection of the engine prior to Dyno testing

SENSOR TIGHTENING TORQUE

Sensor name	Tightening torque
Ne speed	4.9 – 7.9 Nm {0.5 – 0.8 kgm}
Fuel temperature	23 ± 3 Nm {2.35 ± 0.3 kgm}
Coolant temperature	23 ± 3 Nm {2.35 ± 0.3 kgm}
Engine oil pressure	23 ± 3 Nm {2.35 ± 0.3 kgm}
Fuel pressure	Never disassemble.
Intake air temperature	23 ± 3 Nm {2.35 ± 0.3 kgm}
Engine oil temperature	23 ± 3 Nm {2.35 ± 0.3 kgm}

Reference : Factory Standard Reference

Angle tightening bolt	Imposible reuse (replace with new part)
Main Cap	5 punch marks
Connecting rod cap	4 punch marks
Rear Idler Gear	5 punch marks
Fly Wheel	5 punch marks
Cylinder Head	3 punch marks
Injector holder	Lenth below head : 80 mm or more

Reference : Shop Manual Page. 13-23

NOTES :

INSPECTION BY

INSPECTION & APPROVED BY

(MECHANIC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

QR06-MSR-001-

REV. 01

PT.SAPTA INDRA SEJATI

PLANT REBUILD CENTER



**CRANKSHAFT MEASURING
AND INSPECTION CHECKSHEET
KOMATSU ENGINE 6D170E-5**

MACHINE MODEL

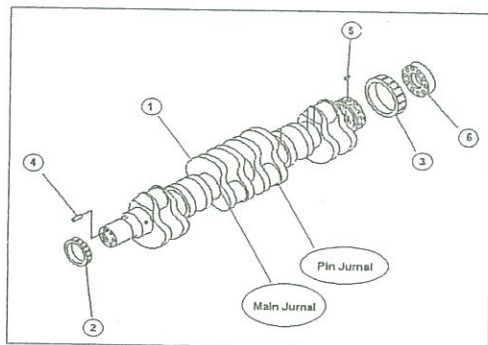
WO NO. :

ENGINE SECTION

CRANK SHAFT MEASURING PROCESS AND INSPECTION CHECKSHEET(AFTER REPAIR)

WO NUMBER		PART NAME	CRANKSHAFT	DOCUMENT NO.	
ENGINE MODEL	6D170E - 5	PART NUMBER	6245-31-1100	PUBL. DATE	16/04/2010
ENGINE S/N		START DATE / TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE / TIME		EKS UNIT CODE	1 of 2

SKETCH DRAWING



Physical Data

No.	Part Name	Qty	New	Reuse
1	Crankshaft	1		
2	Gear	1		
3	Gear	1		
5	Key	1		
6	Spacer	1		

MECHANIC NAME

Standard size

Unit : mm

SIZE	MAIN JOURNAL		PIN JOURNAL	
	STD SIZE	REPAIR LIMIT	STD SIZE	REPAIR LIMIT
STANDARD	140.000 ~ 139.975	139.91	108.000 ~ 107.978	107.91
Under Size 0.25	139.750 ~ 139.725	139.66	107.750 ~ 107.728	107.66
Under Size 0.50	139.500 ~ 139.475	139.41	107.500 ~ 107.478	107.41
ROUNDNESS	0 ~ 0.010	0 ~ 0.015	0 ~ 0.010	0 ~ 0.015

STRAIGHTNESS

MAXIMUM = 0.15 mm (TIR)

FILLET RADIUS CHECK

OK

NOT

Main Journal = 6.0 ~ 6.5 mm

Pin Journal = 6.0 ~ 6.5 mm

VISUAL CHECK

OK

NOT

CRACK HAS BEEN CHECKED

MANHOURS UTILIZATION

	START	FINISH	MAN HOURS
Measuring Process			
TOTAL			

INSPECTION PROCESS

CRITICAL INSPECTION			CHECK POINT (DIMENSION)					ROUNDNESS MAX.
			1			2		
			X - X	Y - Y		X - X	Y - Y	X - Y
MAIN JOURNAL	1	MEC						
		QA						
	2	MEC						
		QA						
	3	MEC						
		QA						
	4	MEC						
		QA						
	5	MEC						
		QA						
	6	MEC						
		QA						
	7	MEC						
		QA						

CRANK SHAFT MEASURING PROCESS AND INSPECTION CHECKSHEET(AFTER REPAIR)

WO NUMBER		PART NAME	CRANKSHAFT	DOCUMENT NO.	
ENGINE MODEL	6D170E - 5	PART NUMBER	6245-31-1100	PUBL. DATE	16/04/2010
ENGINE S/N		START DATE / TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE / TIME		EKS UNIT CODE	2 of 2

CRITICAL INSPECTION			CHECK POINT (DIMENSION)				ROUNDNESS MAX.	
			1		2			
			X - X	Y - Y	X - X	Y - Y		X - Y
PIN JOURNAL	1	MEC						
		QA						
	2	MEC						
		QA						
	3	MEC						
		QA						
	4	MEC						
		QA						
	5	MEC						
		QA						
	6	MEC						
		QA						
SRTAIGHTNESS	MAXIMUM =		mm / TIR =		mm			

CRITICAL INSPECTION		CHECK POINT DIMENSION (LENGTH/WIDTH)						
		1	2	3	4	5	6	7
MAIN JOURNAL	STD SIZE	64.00 ~ 64.50	64.00 ~ 64.50	64.00 ~ 64.50	64.00 ~ 64.50	64.00 ~ 64.50	64.00 ~ 64.05	64.00 ~ 64.50
	QA							
PIN JOURNAL	STD SIZE	72.000~ 72.074	72.000~ 72.074	72.000~ 72.074	72.000~ 72.074	72.000~ 72.074	72.000~ 72.074	
	QA							

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVAL BY :

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)



SERVICES

PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

ENGINE ASS'Y

DELIVERY INSPECTION

BEFORE DYNOTEST

CHECKSHEET

KOMATSU ENGINE 6D170-5 SERIES

MACHINE MODEL

WO NO:

ENGINE ASS'Y SECTION

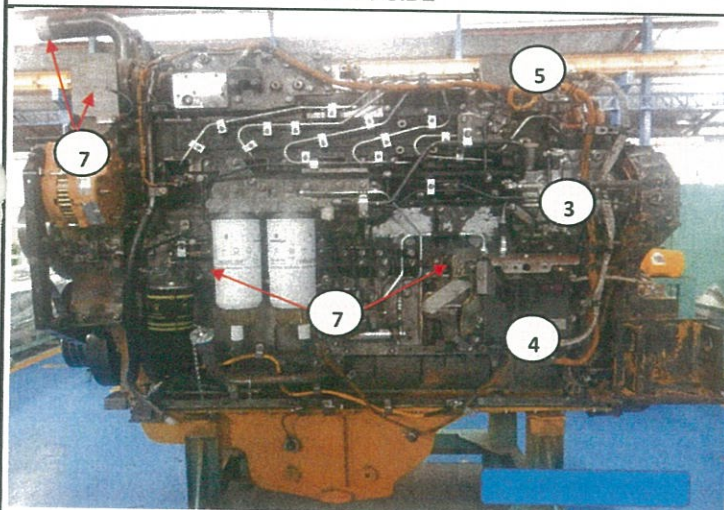
DELIVERY INSPECTION BEFORE DYNOTEST ENGINE CHECKSHEET

DOCUMENT NO. RC/QA/EG/418
PUBL. DATE
REVISION NO.
PAGE

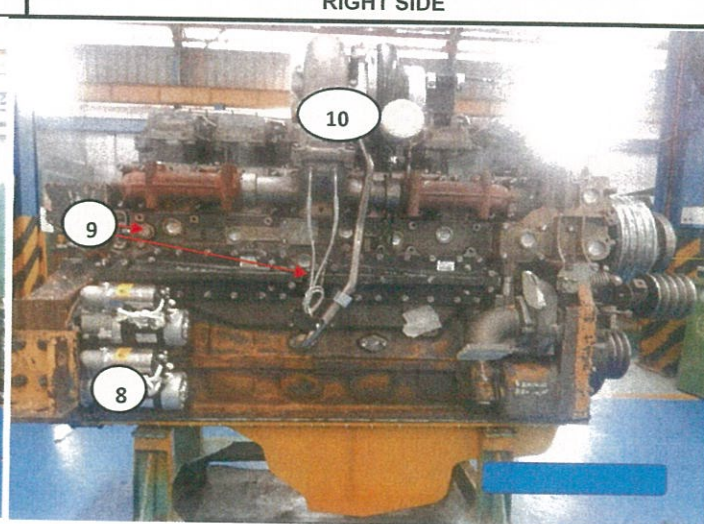
JOB NUMBER		PART NAME	ENGINE ASS'Y
ENGINE MODEL	SAA6D170-5	DATE / TIME	
ENGINE SER. NO			
MACHINE MODEL	HD 465-7R		

SKETCH DRAWING OR PHOTOES

LEFT SIDE



RIGHT SIDE



✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

L H				R H			
NO	ITEM	✓	CONDITION	NO	ITEM	✓	CONDITION
1	Completeness All komponent Engine			8	2 x Starting motor(7,5 KW)		
	-Sub Component			9	All plug		
	-Sensor Engine			10	Turbocharger(KTR 130)		
	-Wiring harness ,Tube and Pipe			11	All torque bolt mounting component(Marking with Permanent Marker)		
2	All torque bolt mounting component(Marking with Permanent Marker)						
3	Fuel Supply Pump Assy						
4	Engine Controler						
5	All Mounting Clam Harnest						
6	all plug (blind plug and tread plug)						
7	all hole should be closed(Tape)						

DELIVERY INSPECTION BEFORE DYNOTEST ENGINE CHECKSHEET

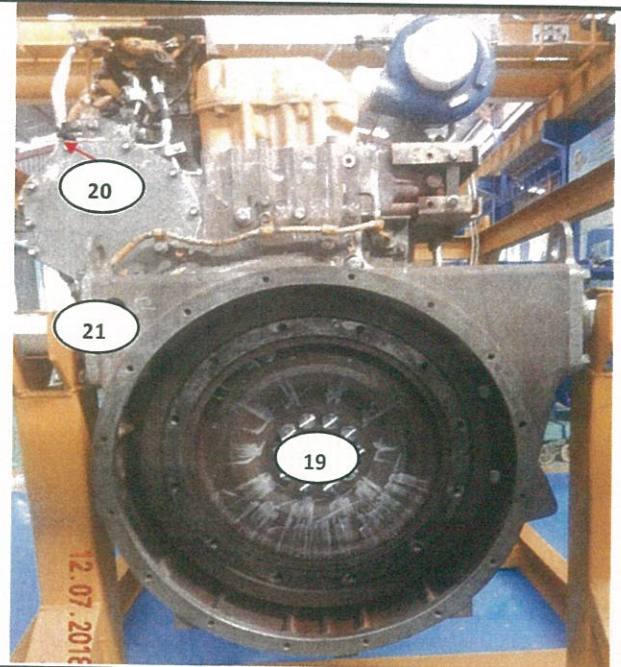
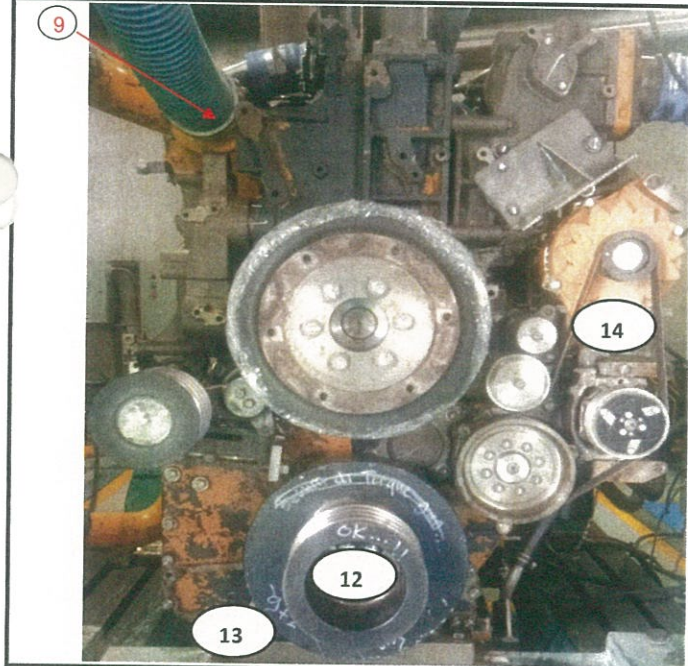
DOCUMENT NO. RC/QA/EG/418
PUBL. DATE
REVISION NO.
PAGE

JOB NUMBER		PART NAME	ENGINE ASS'Y
ENGINE MODEL	SAA6D170-5	DATE / TIME	
ENGINE SER. NO			
MACHINE MODEL	HD 465-7R		

SKETCH DRAWING OR PHOTOES

FRONT SIDE

REAR SIDE



✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

FRONT

REAR

NO	ITEM	✓	INSPECTION CONDITION	NO	ITEM	✓	INSPECTION CONDITION
12	Mounting engine front damper			19	Mounting Flywheel.		
13	Engine front support.			20	All torque bolt mounting component(Marking with Permanent Marker).		
14	Belt alternator(setting Condition)			21	Engine rear support.		
15	All torque bolt mounting component(Marking with Permanent Marker)			22	Completed all checksheet engine		
16	All Mounting Clam Harness						
17	all plug (blind plug and tread plug)						
18	all hole should be closed(Tape)						

DELIVERY BY MAINLINE SECTION

DELIVERED DATE :
MECHANIC NAME :
SIGN

RECEIVING INSPECTION BY DYNO TEST SECTION

RECEIVED DATE :
MECHANIC NAME :
SIGN



PLANT REBUILD CENTER

ENGINE ASS'Y

ENGINE PERFORMANCE

TEST CHECKSHEET

KOMATSU ENGINE 6D170E-5

MACHINE MODEL

D 375-6

JOB NO:

ENGINE ASS'Y SECTION

ENGINE PERFORMANCE TEST REPORT

DOCUMENT NO.	QR05-ENG-011E
PUBLICATION DATE	01/08/2008
REVISION NO.	01
PAGE	

JOB NUMBER		START DATE / TIME	
ENGINE MODEL	SAA6D170E-5	FINISH DATE / TIME	
ENGINE SER.NO.		DYNO.ARM LENGTH	716 mm
MACHINE MODEL	D375A-6 / D375-6R	TEST BY	

RUNNING TEST

No	STANDARD (MANUAL)		ACTUAL TESTED	
	RPM	HP	RPM	HP
1	750	0		
2	1000	32		
3	1200	83		
4	1200	177		
5	1500	330		
6	1800	525		

MEASUREMENT BLOWBY PRESSURE AT RATED SPEED (WITH MANUAL TEST)

	STANDARD MAXIMUM	ACTUAL		REMARK
	350 mmH ₂ O (3.43 Kpa)	BY WATER		
		BY PRESSURE GAUGE		
	ALTERNATOR TEST	STANDARD		ACTUAL
	MEDIUM IDLE :	27.5 ~ 29.5 VDC		

MAINTENANCE STANDARD

SPEED (Rpm)		OIL		WATER		BOOST PRESS (mmHg)	BLOW BY (mmH2O)	EXH TEMP (°C)
		Press. (Kg/cm²)	Temp (°C)	Press. (Kg/cm²)	Temp (°C)			
Low idle	750 ± 50	Min. 1.2	80 ~ 110	-	70~90	Min.950 at rated HP Permissible value 1000	Max 400 permissible 800	Max 700 at rated HP and torque
High idle	2100 ± 40	4.0 ~ 5.5						
POWER (HP) / RPM		524 ~ 606 / 1800 ± 50						
TORQUE (Kgm) / RPM		270 ~ 317/ 1300 ± 100						



PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ENGINE PERFORMANCE TEST REPORT

DOCUMENT NO.

PUBLICATION DATE

01/09/2009

REVISION NO.

01

PAGE

JOB NUMBER		START DATE / TIME	
ENGINE MODEL	SAA6D170E-5	FINISH DATE / TIME	
ENGINE SER.NO.		DYNO.ARM LENGTH	716 mm
MACHINE MODEL	D375-6/D375-6R	TEST BY	

QUALITY PROBLEM

DESCRIPTION

REMARKS

WHAT (PROBLEM NAME)

WHERE (PROBLEM LOCATION)

WHEN (PROBLEM DATE)

WHO (INSPECTED BY)

HOW (PROBLEM EXPLAIN)

CORRECTIVE ACTION :

PREVENTIVE ACTION :

REMARKS :



PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ENGINE PERFORMANCE TEST REPORT	PUBLICATION DATE	
	REVISION NO.	01
	PAGE	

WO NUMBER		START DATE / TIME	
ENGINE MODEL	SAA6D170-5	FINISH DATE / TIME	
ENGINE SER.NO.		DYNO.ARM LENGTH	716 mm
MACHINE MODEL	D 375-6R	TEST BY	

RECHECK VALVE CLEARANCE AFTER LOAD

Adjustment valve clearance	* Penentuan posisi top cylinder engine harus dicek dan disetujui oleh QA. Step and method adjustment of value clearance following to the shop manual attached. Firing Order: 1-5-3-6-2-4 Valve clearance:			<table border="1"> <tr> <th colspan="2">PARAF</th> </tr> <tr> <th>MEC</th> <th>QA</th> </tr> </table>		PARAF		MEC	QA
	PARAF								
	MEC	QA							
	Intake valve : 0.32 ± 0.02	Act :	mm						
Exhaust valve : 0.62 ± 0.02	Act :	mm							
	Tighten locknut to secure adjustment screw. Locknut: 5,9 ~ 7,9 kgm			Act :	kgm				
Rocker Arm cover	Fit O ring and install rocker arm cover. Mounting bolt cover 1,3,4,5&6 = $1 \pm 0,1$ Kgm			Act :	kgm				
	Mounting bolt cover 2 = $6 \sim 7,5$ Kgm			Act :	kgm				

INSPECTION BY

INSPECTION & APPROVAL BY

MECHANIC

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

REV.01



PLANT REBUILD CENTER

ENGINE ASS'Y DELIVERY AND INSPECTION CHECK SHEET

KOMATSU ENGINE 6D170-5

MACHINE MODEL

D 375-6

JOB NO:

ENGINE ASS'Y SECTION

DELIVERY INSPECTION SHEET ENGINE

MACHINE MODEL	D375-6R	ENGINE PART NO.	6240-B0-0040	DOCUMENT NO	RC/QR06/EG/24/07
ENGINE MODEL	SAA6D170E-5	WO NO.		LIFETIME	
ENGINE SERIAL NO		RECEIVING DATE		REVISION NO.	01
HISTORICAL		SITE		PAGE	1 OF 2

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

[illegible]

Notes :